

TryHackMe Journal - Gerry Stevens

Entry 1

Room Name: Linux Fundamentals 1

Date Completed: 02/03/2026

Notes During the Room:

"Linux" is based on UNIX (another operating system) that is an open-source, variant of Linux. Basic functions are needed to be able to do basic functions like navigate to files, output their contents and make files! Command to start with I have broken down in the table below:

Command	Description
echo	Output any text that we provide
whoami	Find out what user we're currently logged in as!
Command	Full Name
ls	listing

<u>cd</u>	change directory
cat	concatenate
pwd	print working directory

Symbol / Operator	Description
&	This operator allows commands to run in the background of the terminal.
&&	This operator allows combining multiple commands together in one line of terminal.
>	This operator is a redirector - meaning that we can take the output from a command (such as using cat to output a file) and direct it elsewhere.
>>	This operator does the same function of the > operator but appends the output rather than replacing (meaning nothing is overwritten).

Important Takeaways:

- Linux is an OS, like Windows. There are many different versions of Linux that serve different purposes.
- Linux systems rely more heavily on the command line to do tasks, like navigate the file system.
- Same basic commands while working with files are ls, cd, cat and pwd

Entry 2

Room Name: Linux Fundamentals 2

Date Completed: 02/05/2026

Notes During the Room:

Commands allow for arguments to be provided identified by a hyphen and a certain keyword known as flags or switches. Using a command, unless otherwise specified, it will perform its default behaviour.

For example, `ls` lists the contents of the working directory, but hidden files are not shown. Flags and switches extend the behaviour of commands. Using `ls` example, `ls` informs that there is only one folder named "folder1" as highlighted in the screenshot below. Note that the contents in the screenshots below are only examples.

```
tryhackme@linux2:~$ ls
folder1
tryhackme@linux2:~$
```

However, after using the `-a` argument (short for `--all`), we now suddenly have an output with a few more files and folders such as ".hiddenfolder". Files and folders with "." are hidden files.

```
tryhackme@linux2:~$ ls -a
.hiddenfolder folder1
tryhackme@linux2:~$
```

Important Takeaways:

This room is theory-heavy. Covers a range of fundamentals to get me familiar with [Linux](#).

This room taught:

- How to connect to a [Linux](#) machine remotely using [SSH](#)
- How to use commands by providing flags, switches
- Other commands to interact with the filesystem and its contents
- A brief introduction to file permissions & switching users

I will be going through this room again once or twice to gain more familiarity with the concepts.

Entry 3

Room Name: Linux Fundamentals 3

Date Completed: 02//15/2026

Notes During the Room:

Introducing terminal text editors:

Only stored text in files using a combination of the `echo` command and the pipe operators (`>` and `>>`) and is not an efficient way to handle data working with files with multiple lines and the sorts. Options to use are a variety of utilities such as `nano`.

Nano

Create or edit a file using nano use `nano filename` -- replacing "filename" with the name of the file to edit. Nano has a few features that are easy to remember & covers the most general things wanted out of a text editor, including:

- Searching for text
- Copying and Pasting
- Jumping to a line number
- Finding out what line number is on

Downloading Files (Wget)

A pretty fundamental feature of computing is the ability to transfer files to download a program, a script, or even a picture and there are multiple ways in which we can retrieve these files. Covering the use of `wget`, this command to download files from the web via HTTP—simply needs to provide the address of the resource to download. For example, to download a file named "myfile.txt" onto machine, assuming web address is known to be something like:

```
wget https://assets.tryhackme.com/additional/linux-fundamentals/part3/myfile.txt
```

Important Takeaways:

The Linux module created more familiarity with Linux to interact with having potential to do very powerful things with relative ease with the introduction to:

- Using terminal text editors
- General utilities such as downloading and serving contents using a python webserver
- A look into processes
- Maintaining & automating system by the use of crontabs, package management, and reviewing logs

Entry 4

Room Name: Intro to Logs

Date Completed: 02/22/2026

Notes During the Room:

Learning Objectives: This room covers how logs can be used to record an adversary's actions, the tools and techniques needed to perform log analysis, and the significance of effectively collecting and analysing logs.

- Understand the importance of logs as a historical activity record for identifying and mitigating potential threats
- Explore various types of logs, logging mechanisms and collection methods across multiple platforms
- Gain hands-on experience detecting and defeating adversaries through log analysis

The True Power of Logs: Contextual Correlation: A single log entry may seem insignificant on its own. But when log data is aggregated, analysed, and cross-referenced with other sources of information, it becomes a potent investigation tool. Logs can answer critical questions about an event, such as:

- What happened?
- When did it happen?
- Where did it happen?
- Who is responsible?
- Were their actions successful?
- What was the result of their action?

Log Types: Specific log types can offer a unique perspective on a system's operation, performance, and security. While there are various log types, we will focus on the most common ones that cover approximately 80% of the typical use cases. Below is a list of some of the most common log types:

- **Application Logs:** Messages about specific applications, including status, errors, warnings, etc.
- **Audit Logs:** Activities related to operational procedures crucial for regulatory compliance.
- **Security Logs:** Security events such as logins, permissions changes, firewall activity, etc.
- **Server Logs:** Various logs a server generates, including system, event, error, and access logs.
- **System Logs:** Kernel activities, system errors, boot sequences, and hardware status.
- **Network Logs:** Network traffic, connections, and other network-related events.
- **Database Logs:** Activities within a database system, such as queries and updates.
- **Web Server Logs:** Requests processed by a web server, including URLs, response codes, etc.

Log Formats: A log format defines the structure and organisation of data within a log file. It specifies how the data is encoded, how each entry is delimited, and what fields are included in each row. These formats can vary widely and may fall into three main categories: Semi-structured, Structured, and Unstructured.

Entry 4 [continued]

Log Collection: Log collection is an essential component of log analysis, involving the aggregation of logs from diverse sources such as servers, network devices, software, and databases.

For logs to effectively represent a chronological sequence of events, it's crucial to maintain the system's time accuracy during logging. Utilising the Network Time Protocol (NTP) is a method to achieve this synchronisation and ensure the integrity of the timeline stored in the logs. As this is a foundational step to ensuring that a security analyst would have a comprehensive data set to review, the following is a simple step-by-step process to achieving this, bearing in mind the need to prioritise the collection based on significant information:

- **Identify Sources:** List all potential log sources, such as servers, databases, applications, and network devices.
- **Choose a Log Collector:** Opt for a suitable log collector tool or software that aligns with the infrastructure.
- **Configure Collection Parameters:** Ensure that time synchronisation is enabled through NTP to maintain accurate timelines, adjust settings to determine which events to log at what intervals, and prioritise based on importance.
- **Test Collection:** Once configured, run a test to ensure logs are appropriately collected from all sources.

Log Management: Efficient Log Management ensures that every gathered log is stored securely, organised systematically, and is ready for swift retrieval. A hybrid approach can provide a balanced solution by hoarding all log files and selectively trimming. Once collating of logs, effective management of them is paramount. These steps can be followed to achieve this:

- **Storage:** Decide on a secure storage solution, considering factors like retention period and accessibility.
- **Organisation:** Classify logs based on their source, type, or other criteria for easier access later.
- **Backup:** Regularly back up logs to prevent data loss.
- **Review:** Periodically review logs to ensure they are correctly stored and categorised.

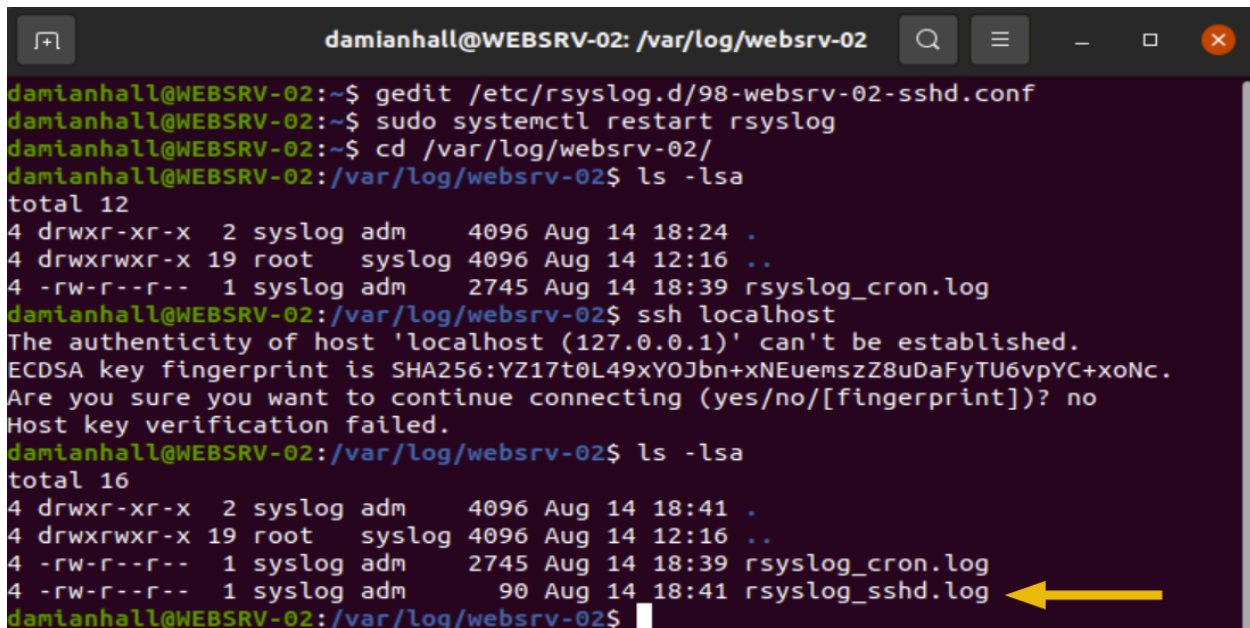
Log Centralization: Centralization is pivotal for swift log access, in-depth analysis, and rapid incident response. A unified system allows for efficient log management with tools that offer real-time detection, automatic notifications, and seamless integration with incident management systems. Centralizing logs can significantly streamline access and analysis. Here's a simple process for achieving it:

- **Choose a Centralised System:** Opt for a system that consolidates logs from all sources, such as the Elastic Stack or Splunk.
- **Integrate Sources:** Connect all log sources to this centralised system.
- **Set Up Monitoring:** Utilise tools that provide real-time monitoring and alerts for specified events.
- **Integration with Incident Management:** Ensure that a centralized system can integrate seamlessly with any incident management tools or protocols in place.

Entry 4 [continued]

Practical Activity: Log Collection with rsyslog - This activity aims to introduce **rsyslog** and demonstrate how it can enhance the centralization and management of logs. As part of the collection process, we will configure **rsyslog** to log all **sshd** messages to a specific file, such as **/var/log/websrv-02/rsyslog_sshd.log**. The steps below can be followed to achieve this:

1. **Open a Terminal.**
2. **Ensure rsyslog is Installed:** Check if **rsyslog** is installed by running the command: **sudo systemctl status rsyslog**
3. **Create a Configuration File:** Use a text editor to create the following configuration file: **gedit /etc/rsyslog.d/98-websrv-02-sshd.conf**, **nano /etc/rsyslog.d/98-websrv-02-sshd.conf**, **vi /etc/rsyslog.d/98-websrv-02-sshd.conf**, or **vim /etc/rsyslog.d/98-websrv-02-sshd.conf**
4. **Add the Configuration:** Add the following lines in **/etc/rsyslog.d/98-websrv-02-sshd.conf** to direct the **sshd** messages to the specific log file:
\$FileCreateMode 0644
:programname, isequal, "sshd" /var/log/websrv-02/rsyslog_sshd.log
5. **Save and Close the Configuration File.**
6. **Restart rsyslog:** Apply the changes by restarting **rsyslog** with the command: **sudo systemctl restart rsyslog**
7. **Verify the Configuration:** Verify the configuration works by initiating an SSH connection to localhost via **ssh localhost** or by checking the log file after a minute or two.



```
damianhall@WEBSRV-02: /var/log/websrv-02
damianhall@WEBSRV-02:~$ gedit /etc/rsyslog.d/98-websrv-02-sshd.conf
damianhall@WEBSRV-02:~$ sudo systemctl restart rsyslog
damianhall@WEBSRV-02:~$ cd /var/log/websrv-02/
damianhall@WEBSRV-02:/var/log/websrv-02$ ls -lsa
total 12
4 drwxr-xr-x  2 syslog adm    4096 Aug 14 18:24 .
4 drwxrwxr-x 19 root  syslog 4096 Aug 14 12:16 ..
4 -rw-r--r--  1 syslog adm    2745 Aug 14 18:39 rsyslog_cron.log
damianhall@WEBSRV-02:/var/log/websrv-02$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ECDSA key fingerprint is SHA256:YZ17t0L49xY0Jbn+xNEuemszZ8uDaFyTU6vpYC+xoNc.
Are you sure you want to continue connecting (yes/no/[fingerprint])? no
Host key verification failed.
damianhall@WEBSRV-02:/var/log/websrv-02$ ls -lsa
total 16
4 drwxr-xr-x  2 syslog adm    4096 Aug 14 18:41 .
4 drwxrwxr-x 19 root  syslog 4096 Aug 14 12:16 ..
4 -rw-r--r--  1 syslog adm    2745 Aug 14 18:39 rsyslog_cron.log
4 -rw-r--r--  1 syslog adm     90 Aug 14 18:41 rsyslog_sshd.log
damianhall@WEBSRV-02:/var/log/websrv-02$
```

A yellow arrow points to the newly created file **rsyslog_sshd.log** in the second **ls -lsa** output.

Entry 4 [continued]

Log Storage: Logs can be stored in various locations, such as the local system that generates them, a centralised repository, or cloud-based storage. The choice of storage location typically depends on multiple factors:

- **Security Requirements:** Ensuring that logs are stored in compliance with organisational or regulatory security protocols.
- **Accessibility Needs:** How quickly and by whom the logs need to be accessed can influence the choice of storage.
- **Storage Capacity:** The volume of logs generated may require significant storage space, influencing the choice of storage solution.
- **Cost Considerations:** The budget for log storage may dictate the choice between cloud-based or local solutions.
- **Compliance Regulations:** Specific industry regulations governing log storage can affect the choice of storage.
- **Retention Policies:** The required retention time and ease of retrieval can affect the decision-making process.
- **Disaster Recovery Plans:** Ensuring the availability of logs even in system failure may require specific storage solutions.

Log Retention: It is vital to recognise that log storage is not infinite. Therefore, a reasonable balance between retaining logs for potential future needs and the storage cost is necessary. Understanding the concepts of Hot, Warm, and Cold storage can aid in this decision-making:

- **Hot Storage:** Logs from the past 3-6 months that are most accessible. Query speed should be near real-time, depending on the complexity of the query.
- **Warm Storage:** Logs from six months to 2 years, acting as a data lake, easily accessible but not as immediate as Hot storage.
- **Cold Storage:** Archived or compressed logs from 2-5 years. These logs are not easily accessible and are usually used for retroactive analysis or scoping purposes.

Log Deletion: Log deletion must be performed carefully to avoid removing logs that could still be of value. The backup of log files, especially crucial ones, is necessary before deletion. It is essential to have a well-defined deletion policy to ensure compliance with data protection laws and regulations.

Log deletion helps to:

- Maintain a manageable size of logs for analysis.
- Comply with privacy regulations, such as GDPR, which require unnecessary data to be deleted.
- Keep storage costs in balance.

Entry 4 [continued]

Best Practices: Log Storage, Retention and Deletion:

- Determine the storage, retention, and deletion policy based on both business needs and legal requirements.
- Regularly review and update the guidelines per changing conditions and regulations.
- Automate the storage, retention, and deletion processes to ensure consistency and avoid human errors.
- Encrypt sensitive logs to protect data.
- Regular backups should be made, especially before deletion.

Practical Activity: Log Management with logrotate - This activity aims to introduce **logrotate**, a tool that automates log file rotation, compression, and management, ensuring that log files are handled systematically. It allows automatic rotation, compression, and removal of log files. As an example, here's how we can set it up for **/var/log/websrv-02/rsyslog_sshd.log**:

1. **Create a Configuration File:** `sudo gedit /etc/logrotate.d/98-websrv-02_sshd.conf`, `sudo nano /etc/logrotate.d/98-websrv-02_sshd.conf`, `sudo vi /etc/logrotate.d/98-websrv-02_sshd.conf`, or `sudo vim /etc/logrotate.d/98-websrv-02_sshd.conf`
2. **Define Log Settings:**

```
/var/log/websrv-02/rsyslog_sshd.log {
    daily
    rotate 30
    compress
    lastaction
    DATE=$(date +"%Y-%m-%d")
    echo "$(date)" >> "/var/log/websrv-02/hashes_"$DATE"_rsyslog_sshd.txt"
    for i in $(seq 1 30); do
        FILE="/var/log/websrv-02/rsyslog_sshd.log.$i.gz"
        if [ -f "$FILE" ]; then
            HASH=$(/usr/bin/sha256sum "$FILE" | awk '{ print $1 }')
            echo "rsyslog_sshd.log.$i.gz "$HASH"" >> "/var/log/websrv-02/hashes_"$DATE"_rsyslog_sshd.txt"
        fi
    done
    systemctl restart rsyslog
endscript
}
```

3. **Saved and Closed.**
4. **Manual Execution:** `sudo logrotate -f /etc/logrotate.d/98-websrv-02_sshd.conf`

Entry 4 [continued]

```
damianhall@WEBSRV-02: /var/log/websrv-02
damianhall@WEBSRV-02:/var/log/websrv-02$ sudo nano /etc/logrotate.d/98-websrv-02_sshd.conf
damianhall@WEBSRV-02:/var/log/websrv-02$ ls -lsa
total 24
 4 drwxr-xr-x  2 syslog adm    4096 Aug 14 21:39 .
 4 drwxrwxr-x 19 root  syslog 4096 Aug 14 21:36 ..
12 -rw-r--r--  1 syslog adm   9911 Aug 14 21:40 rsyslog_cron.log
 4 -rw-r--r--  1 syslog adm    90 Aug 14 21:39 rsyslog_sshd.log
damianhall@WEBSRV-02:/var/log/websrv-02$ sudo logrotate -f /etc/logrotate.d/98-websrv-02_sshd.conf
damianhall@WEBSRV-02:/var/log/websrv-02$ ls -lsa
total 28
 4 drwxr-xr-x  2 syslog adm    4096 Aug 14 21:40 .
 4 drwxrwxr-x 19 root  syslog 4096 Aug 14 21:36 ..
 4 -rw-r--r--  1 root   root   119 Aug 14 21:40 hashes_2023-08-14_rsyslog_sshd.txt
12 -rw-r--r--  1 syslog adm   9911 Aug 14 21:40 rsyslog_cron.log
 4 -rw-r--r--  1 syslog adm    108 Aug 14 21:39 rsyslog_sshd.log.1.gz
damianhall@WEBSRV-02:/var/log/websrv-02$ cat hashes_2023-08-14_rsyslog_sshd.txt
Mon 14 Aug 2023 09:40:27 PM UTC
rsyslog_sshd.log.1.gz 309ad0c79a41765a484c3f07d52502ba0e675e80144ffa76749a544be6bf6e8a
damianhall@WEBSRV-02:/var/log/websrv-02$
```

Important Takeaways:

Congratulations! You've completed the Intro to Logs room.

In summary, we were able to learn and perform the following:

- The significance of logs as records of past activities; essential for pinpointing and tackling threats.
- Delve into an array of logs, their creation techniques, and the methods of gathering them from diverse systems.
- Review the results from analysing logs in the realms of detection engineering and incident handling.
- Acquire practical skills in identifying and countering adversaries via log analysis.

If you enjoyed this room, continue learning and developing proficiency in areas specific to Security Operations and Incident Response tooling, which may enhance your log analysis and overall Blue Teaming skills such as the following:

Entry 5

Room Name: Wireshark Basics

Date Completed: 03/06/2026

Notes During the Room:

Wireshark is an open-source, cross-platform network packet analyser tool capable of sniffing and investigating live traffic and inspecting packet captures (PCAP). It is commonly used as one of the best packet analysis tools. In this room, we will look at the basics of Wireshark and use it to perform fundamental packet analysis.

Completed Tasks:

Loading PCAP Files: From a recently processed `http1.pcap` file loaded to see Wireshark's detailed packet presentation w/ use the "File" menu, dragging and dropping the file, or double-clicking on the file to load a pcap.

Coloring Packets: Color packets in order of different conditions and the protocol to spot anomalies and protocols in captures quickly (this explains why almost everything is green in the given screenshots) helping to track down what is being looked for during analysis.

Traffic Sniffing: Started network sniffing (capturing traffic) using an interface with the status bar to stop sniffing of packets and the restarting of the sniffing process as status bar also interfaces with the number of collected packets.

Merge PCAP Files: Wireshark combines two pcap files into one single file using the "**File → Merge**" menu path to merge a pcap with the processed one while choosing the second file, Wireshark will show the total number of packets in the selected file. Save the "merged" pcap file before working on it.

Packet Dissection: Protocol dissection investigates packet details by decoding available protocols and fields that cover how Wireshark uses OSI layers to break down packets and how to use these layers for analysis.

Packet Details: Clicked on packet in the packet list pane to open its details (double-click will open details in a new window). Packets consist of 5 to 7 layers based on the OSI model. Saw distinct layers to the packet: `frame/packet`, `source [MAC]`, `source [IP]`, `protocol`, `protocol errors`, `application protocol`, and `application data`.

Entry 5 [continued]

Packet Numbers: Wireshark calculates the number of investigated packets. Assigns a unique number for each packet that helps the analysis process for big captures being easy to go back to a specific point event.

Go to Packet: Packet numbers do not only help to count the total number of packets or make it easier to find/investigate specific packets and navigate between packets up and down; it also provides in-frame packet tracking and finds the next packet in the particular part of the conversation.

Find Packets: Apart from packet number, Wireshark can find packets by packet content by using the **"Edit → Find Packet"** menu and help analyze and administer to find specific intrusion patterns or failure traces.

Mark Packets: Marking packets helpful functionality to find/point to a specific packet for further investigation by marking it with use of the **"Edit"** or the **"right-click"** menu to mark/unmark packets showing as Marked packets is shown in black regardless of the original colour by connection type and renewed every file session, so marked packets will be lost after closing the capture file.

Packet Comments: Similar to packet marking, commenting helps as a feature to add comments for particular packets for further investigation or remind and point out important/suspicious points for other layer analysis, and Packet Comments can stay within the capture file until the operator removes them.

Export Packets: Capturing files containing thousands of packets in a single file necessary to separate specific packages from the file to dig deeper to resolve an incident also helped to analyze suspicious packages (decided scope). Use the **"File"** menu to export packets.

Export Objects (Files): Wireshark used to extract files and transfer them through the wire vital to discover shared files and to be saved for further investigation where export objects are for selected protocol streams such as: DICOM, HTTP, IMF, SMB and TFTP.

Time Display Format: Wireshark lists the packets as they are captured that shows time in "Seconds Since Beginning of Capture" with the common usage is the UTC Time Display Format or use the **"View → Time Display Format"** menu to change the time display format.

Expert Info: Wireshark used to detect specific states of protocols to help analyze and spot possible anomalies and problems while understanding that there are chances for false positives/negatives as Expert info provides categories in three different severities.

Packet Filtering: Wireshark's powerful filter engine helps analyze to narrow down the traffic and focus on the event of interest having two types of filtering approaches: capture and display filters. Capture filters are used for **"capturing"** only the packets valid for the used filter. Display filters are used for **"viewing"** the packets valid for the used filter.

Entry 5 [continued]

Apply as Filter: Used as the most basic way of filtering traffic by investigating a captured field is filtered using the "right-click menu" or "**Analyse → Apply as Filter**" menu to filter the specific value. Total and displayed packets are always shown on the status bar. Use of the "**Apply as a Filter**" option filters not only a single entity of the packet but also by investigating a specific packet number and all linked packets by focusing on IP addresses and port numbers.

Conversation Filter: "**Conversation Filter**" option helps view only the related packets and hide the rest of the packets easily. Use the "right-click menu" or "**Analyse → Conversation Filter**" menu to filter conversations.

Colourise Conversation: Option is similar to the "**Conversation Filter**" with one difference as it highlights the linked packets without applying a display filter and decreasing the number of viewed packets in which this option works with the "**Colouring Rules**" option and changes the packet colours without considering the previously applied colour rule that also can use the "**right-click menu**" or "**View → Colourise Conversation**" menu to colourise a linked packet in a single click and use the "**View → Colourise Conversation → Reset Colourisation**" menu to undo this operation.

Prepare as Filter: Similar to "Apply as Filter", this option helps analysts create display filters using the "right-click" menu. However, unlike the previous one, this model doesn't apply the filters after the choice. It adds the required query to the pane and waits for the execution command (enter) or another chosen filtering option by using the "**.. and/or..**" from the "**right-click menu**".

Apply as Column: By default, the packet list pane provides basic information about each packet. Use the "right-click menu" or "**Analyse → Apply as Column**" menu to add columns to the packet list pane. Once clicking on a value and applying it as a column, it will be visible on the packet list pane. This function helps analysts examine the appearance of a specific value/field across the available packets in the capture file. Enable/disable the columns shown in the packet list pane by clicking on the top of the packet list pane.

Follow Stream: Wireshark displays everything in packet portion size but also can reconstruct the streams and view the raw traffic as it is presented at the application level to help analyze and recreate the application-level data of the event of interest. Can view unencrypted protocol data like usernames, passwords and other transferred data while using the "right-click menu" or "**Analyse → Follow TCP/UDP/HTTP Stream**" menu to follow traffic streams. Once a filter is applied, the number of the viewed packets change.

Simple Display Filter Queries: The easiest way to filter quickly the huge amount of packets, is by applying a display filter using the "AppThere are many filter queries available and each of them can be extensively tweaked to show very specific results. Below are some simple filters to get started.

Filter By Protocol Name or Port: There are two basic ways to filter based on a specific protocol: by protocol name and by protocol port number. To filter by **protocol name**, type in the protocol name and enter or click on the arrow button of the display filter bar. Filters shown are for other protocols using keywords like `arp`, `dhcp`, `ftp`, `smtp`, `pop`, `imap`, and more. To filter by **protocol port number**, use the structure "`tcp.port == <port number>`" or "`udp.port == <port number>`". For example, for only `http` packets, use the filter "`tcp.port == 80`" and enter.

Entry 5 [continued]

Filter By IP: When analyzing a packet capture, there is often a need to filter for a specific IP. To filter for a specific IP, use the structure "**ip.addr == <IP address>**". To search for the **IP 192.168.1.2**, filter would be "**ip.addr == 192.168.1.2**".

Important Takeaways: Learned Objectives

- Navigated and configured Wireshark
- Inspected packets and discover information from the different layers of TCP/IP
- Apply display filters

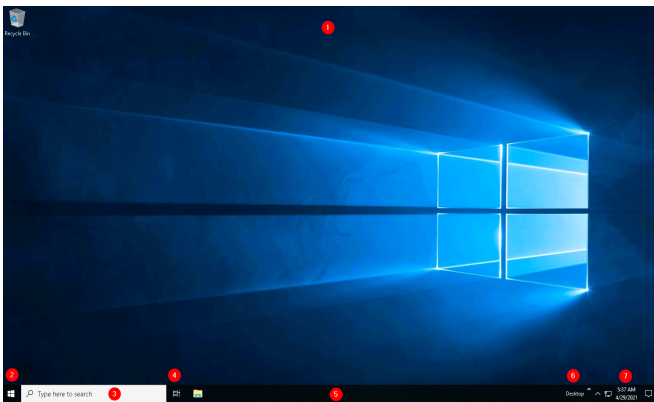
Entry 6

Room Name: Windows Fundamentals 1

Date Completed: 03/15/2026

Notes During the Room:

The Windows operating system history dates back to 1985, and currently. Dominant operating system in both home use and corporate networks & targeted by hackers & malware writers. Today the Windows operating system version for desktop computers is Windows 11 Home and Pro. The current version of the Windows operating system for servers is: Windows Server 2025. Microsoft, have made long strides to improve the usability and security with each new version of Windows.



The above screenshot is an example of a typical Windows Desktop. Each component of GUI is explained:

1. The Desktop
2. Start Menu
3. Search Box (Cortana)
4. Task View
5. Taskbar
6. Toolbars
7. Notification Area

Entry 6 [continued]

The Windows operating system (OS) is a complex product with many system files, utilities, settings, and features. This module will attempt to provide a general overview of just a handful of what makes up the Windows OS, navigate the user interface, and make changes to the system. The content is aimed at those who wish to understand and use the Windows OS on a more comfortable level. User accounts can be one of two types on a typical local Windows system: **Administrator & Standard User**. The user account type will determine what actions the user can perform on that specific Windows system.

- An Administrator can make changes to the system: add users, delete users, modify groups, modify settings on the system, and more.
- A Standard User can only make changes to folders/files attributed to the user & can't perform system-level changes, such as install programs.

There are several ways to determine which user accounts exist on the system. One way is to click the **Start Menu** and type Other User. A shortcut to **System Settings > Other users** should appear. Another way to access this information, and then some, is using Local User and Group Management. Right-click on the Start Menu and click Run. Type **lusrmgr.msc**

Important Takeaways:

As a simple note for a generic overview of the Windows OS. There are intermediate and advanced topics for each topic and/or tasks. Other modules also covered are topics like the Windows folder, the management console, security tools (Windows Defender, Windows Firewall, etc.) and more.

Entry 7

Room Name: Windows Fundamentals 2

Date Completed: 03/22/2026

Notes During the Room:

The **System Configuration** utility (**MSConfig**) is for advanced troubleshooting, and its main purpose is to help diagnose startup issues. Local administrator rights to open this utility have five tabs across the top. Below are the names for each tab. We will briefly cover each tab in this task:

1. General
2. Boot
3. Services
4. Startup
5. Tools

Continuing with Tools available through the **System Configuration** panel. **User Account Control (UAC)** was covered in the **UAC** settings and can be changed or even turned off entirely (not recommended).

Entry 7 [continued]

The slider can be moved to see how the setting will change the UAC settings and Microsoft's stance on the setting which has four security levels, each of which controls how Windows alerts when apps or users try to make changes at the system level falling into four standard categories shown below:

- **Always notify:** This is the highest security. Windows notifies whenever any of the apps have changes made, and the desktop dims (Secure Desktop).
- **Notify for apps:** Windows notifies only when apps try to make changes, but changes to **Windows settings**. This option is enabled by default.
- **Notify without dimming:** Same as above (Notify for apps), but this time the screen does not dim.
- **Never notify:** Notifications are turned off. Windows won't warn about any changes made by any apps.

Find the current level by looking at the position of the slider in the User Account Control settings window, as shown below:



Choose when to be notified about changes to your computer

User Account Control helps prevent potentially harmful programs from making changes to your computer.

[Tell me more about User Account Control settings](#)

Always notify



Never notify

Notify me only when apps try to make changes to my computer (default)

- Don't notify me when I make changes to Windows settings

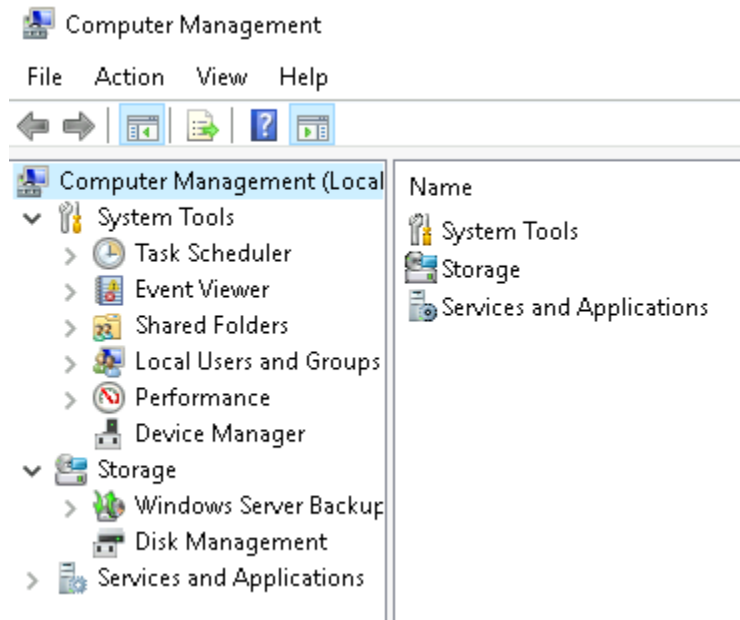
i Recommended if you use familiar apps and visit familiar websites.



Cancel

Entry 7 [continued]

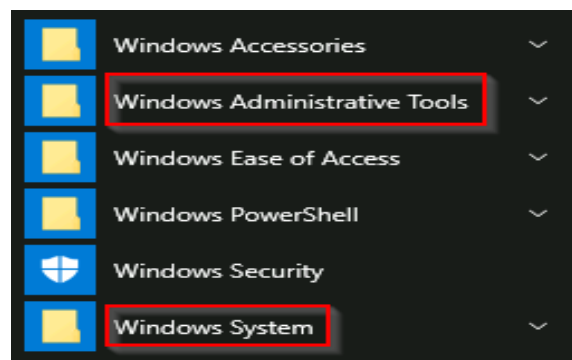
Continuing with tools available through the **System Configuration** panel. The **Computer Management (compmgmt)** utility has three primary sections: **System Tools**, **Storage**, and **Services and Applications**.



- **Always notify:** This is the highest security. Windows notifies whenever any apps are attempted to make changes, and the desktop dims (Secure Desktop).
- **Notify for apps:** Windows notifies only when apps try to make changes, but not when with change in Windows settings. This option is enabled by default.
- **Notify without dimming:** Same as above (Notify for apps), but this time the screen does not dim.
- **Never notify:** Notifications are turned off. Windows won't warn about any changes made on any of the apps. The Computer Management (**compmgmt**) utility has three primary sections: System Tools, Storage, and Services and Applications.

Important Takeaways:

Recall that the tasks covered in this room were some of the tools that can launch from **MSConfig**. Throughout the room, commands and shortcuts were shared for the utilities meaning that no need to launch MSConfig to run these utilities. Can also run some of these utilities directly from the Start Menu. See below where some of these utilities can be found.



Entry 8

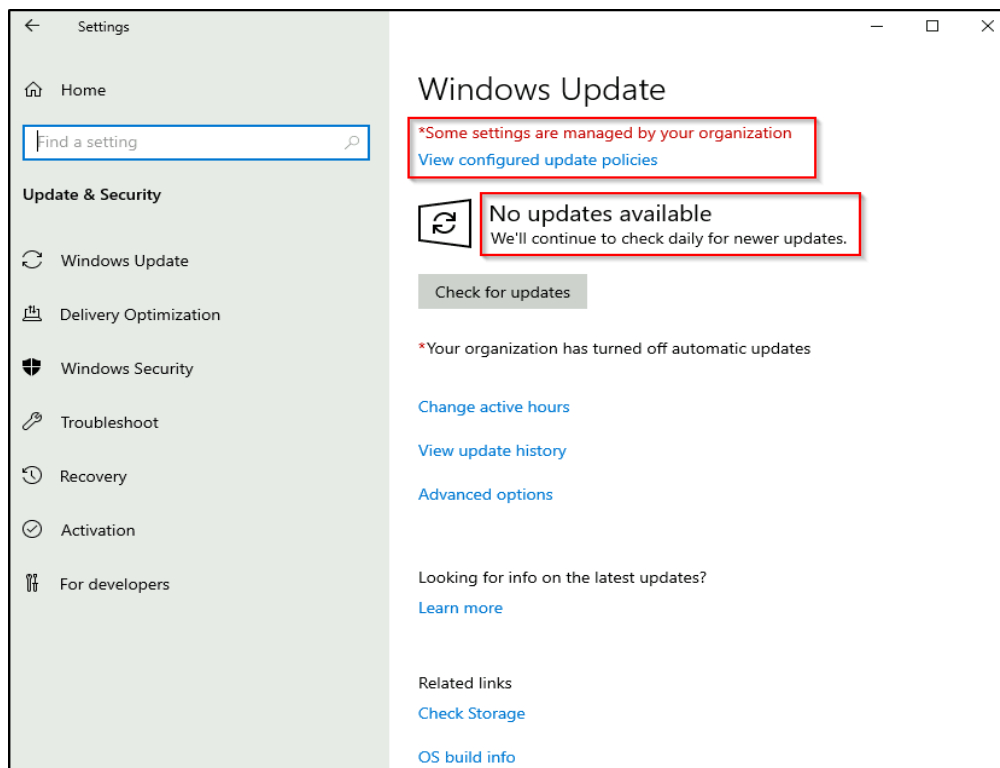
Room Name: Windows Fundamentals 3

Date Completed: 03/25/2026

Notes During the Room:

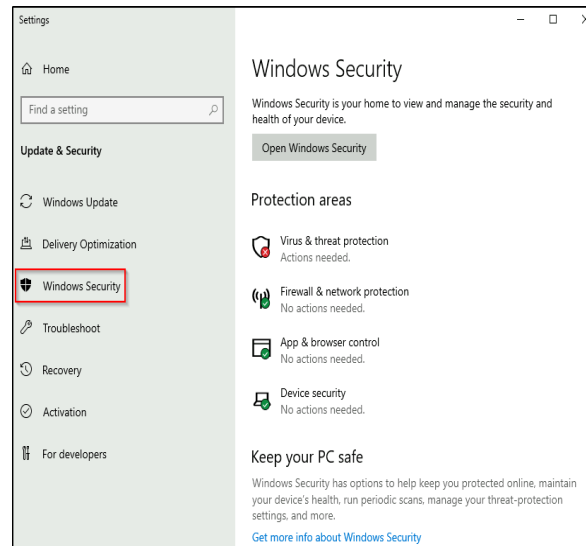
Windows Update is a service provided by Microsoft to provide security updates, feature enhancements, and patches for the Windows operating system and other Microsoft products, such as Microsoft Defender. Updates are typically released on the 2nd Tuesday of each month. This day is called Patch Tuesday. That doesn't necessarily mean that a critical update/patch has to wait for the next Patch Tuesday to be released. If the update is urgent, then Microsoft will push the update via the Windows Update service to the Windows devices. Windows Update is located in Settings. Another way to access Windows Update is from the Run dialog box, or CMD, by running the command `control /name Microsoft.WindowsUpdate`.

1. The **Windows Update** settings are 'managed'. (Typically, home users will not see this type of message)
2. There are no available updates available for the virtual machine. (The attached virtual machine does not have Internet access to communicate with Microsoft to obtain new updates)



Entry 8 [continued]

Per Microsoft, "**Windows Security** is home to manage the tools that protect devices and data". **Windows Security** is also available in Settings.

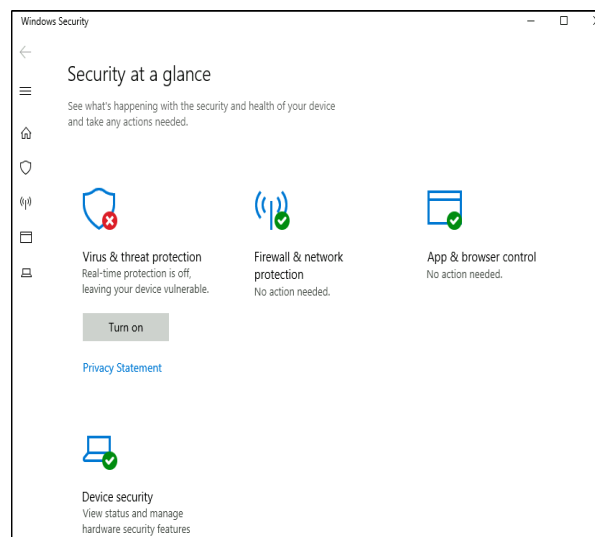


In the above image, focus attention on **Protection areas**.

- **Virus & threat protection**
- **Firewall & network protection**
- **App & browser control**
- **Device security**

After clicking on **Open Windows Security** above provided is a quick comment on the status icons.

- **Green** means the device is sufficiently protected, and there aren't any recommended actions.
- **Yellow** means there is a safety recommendation to review.
- **Red** is a warning that something needs immediate attention.

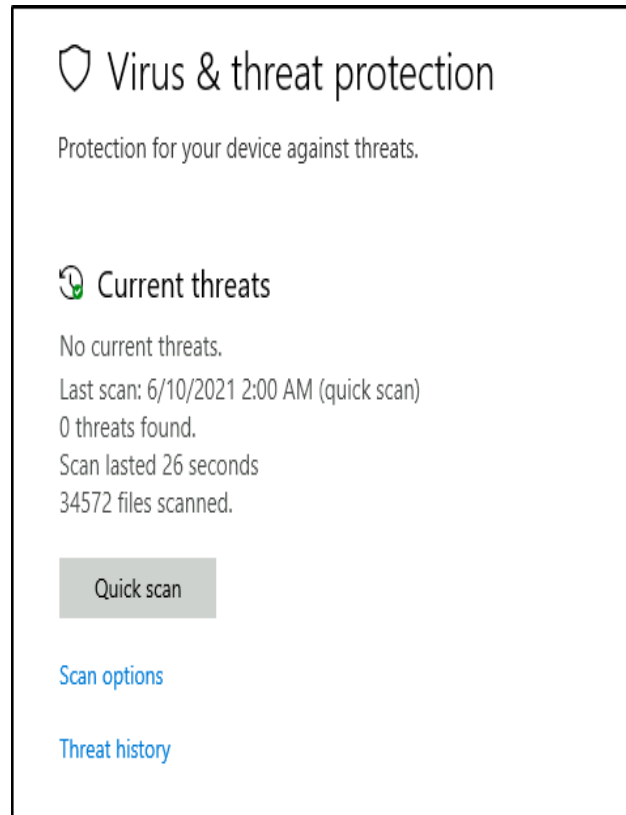


Entry 8 [continued]

Virus & threat protection is divided into two parts:

- **Current threats**
- **Virus & threat protection settings**

The image below only focuses on **Current Threats**.



Current threats

Scan options

- **Quick scan** - Checks folders in systems where threats are commonly found.
- **Full scan** - Checks all files and running programs on the hard disk. This scan could take longer than one hour.
- **Custom scan** - Choose which files and locations wanted to be checked.

Threat history

- **Last scan** - Windows Defender Antivirus automatically scans devices for viruses and other threats to help keep it safe.
- **Quarantined threats** - Quarantined threats have been isolated and prevented from running on a device. They will be periodically removed.
- **Allowed threats** - Allowed threats are items identified as threats, which are allowed to run on a device.

Entry 8 [continued]

Virus & threat protection settings



Virus & threat protection settings

Manage settings

- **Real-time protection** - Locates and stops malware from installing or running on devices.
- **Cloud-delivered protection** - Provides increased and faster protection with access to the latest protection data in the cloud.
- **Automatic sample submission** - Send sample files to Microsoft to help protect others from potential threats.
- **Controlled folder access** - This feature, if enabled, protects files, folders, and memory areas on a device from unauthorized changes by malicious or unknown applications. If it is enabled, only approved and trusted apps would be allowed to modify the files in the protected folders. To enable this feature, click on the **Manage controlled folder access** button under **Controlled Folder Access** and turn it on.
- **Exclusions** - Windows Defender Antivirus allows to exclude any files or folders from the antivirus scanning. This is done to reduce the number of false positives. Administrators might not want the antivirus to scan specific files or folders. By adding them to the exclusions list, the antivirus would ignore them and scan all the other files and folders. To add any file or folder to the Windows Defender exclusion list, click on the **Add or remove exclusions** button under **Exclusions** and add as many exclusions as desired.
- **Notifications** - Windows Defender Antivirus will send notifications with critical information about the health and security of a device.

Entry 8 [continued]

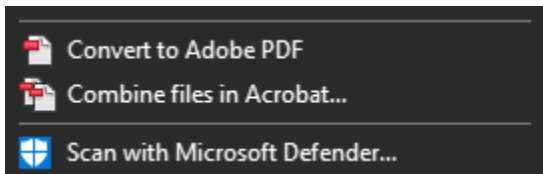
Virus & threat protection updates

- Check for updates - Manually check for updates to update Windows Defender Antivirus definitions.

Ransomware protection

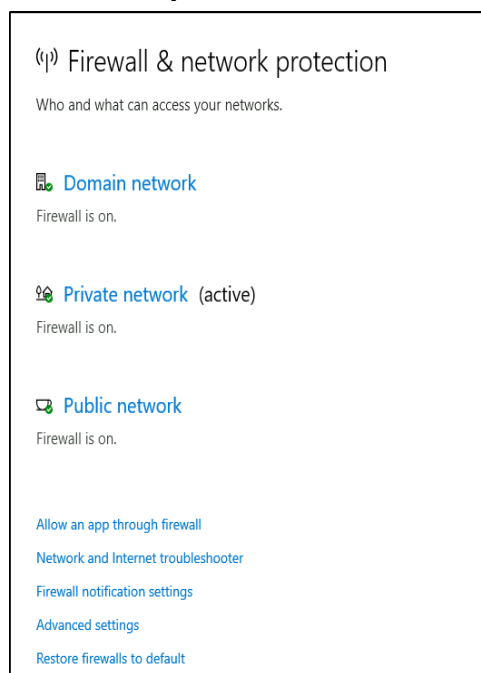
- Controlled folder access - Ransomware protection requires this feature to be enabled, which in turn requires Real-time protection to be enabled.

Note: Real-time protection is turned off in the attached VM to decrease the chances of performance issues. Since the VM can't reach the Internet and there aren't any threats in the VM, this is safe to do. Real-time protection should definitely be enabled in personal Windows devices unless there are 3rd party products that provide the same protection. Ensure it's always up-to-date and enabled. To perform on-demand scans on any file/folder by right-clicking the item and selecting '**Scan with Microsoft Defender**'. The below image was taken from another Windows device to show this feature.



What is a firewall?

Per Microsoft, "Traffic flows into and out of devices via what we call ports. A firewall is what controls what is - and more importantly isn't - allowed to pass through those ports like a security guard standing at the door, checking the ID of everything that tries to enter or exit". The image below will reflect what will be seen to navigate to **Firewall & network protection**.



Entry 8 [continued]

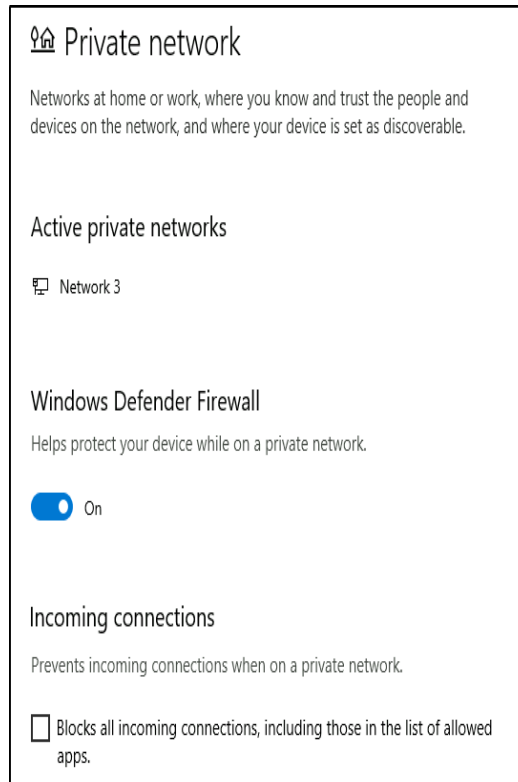
Note: Each network may have different status icons.

What is the difference between the 3 (**Domain**, **Private**, and **Public**)?

Per Microsoft, "Windows Firewall offers three firewall profiles: domain, private and public".

- **Domain** - The domain profile applies to networks where the host system can authenticate to a domain controller.
- **Private** - The private profile is a user-assigned profile and is used to designate private or home networks.
- **Public** - The default profile is the public profile, used to designate public networks such as Wi-Fi hotspots at coffee shops, airports, and other locations.

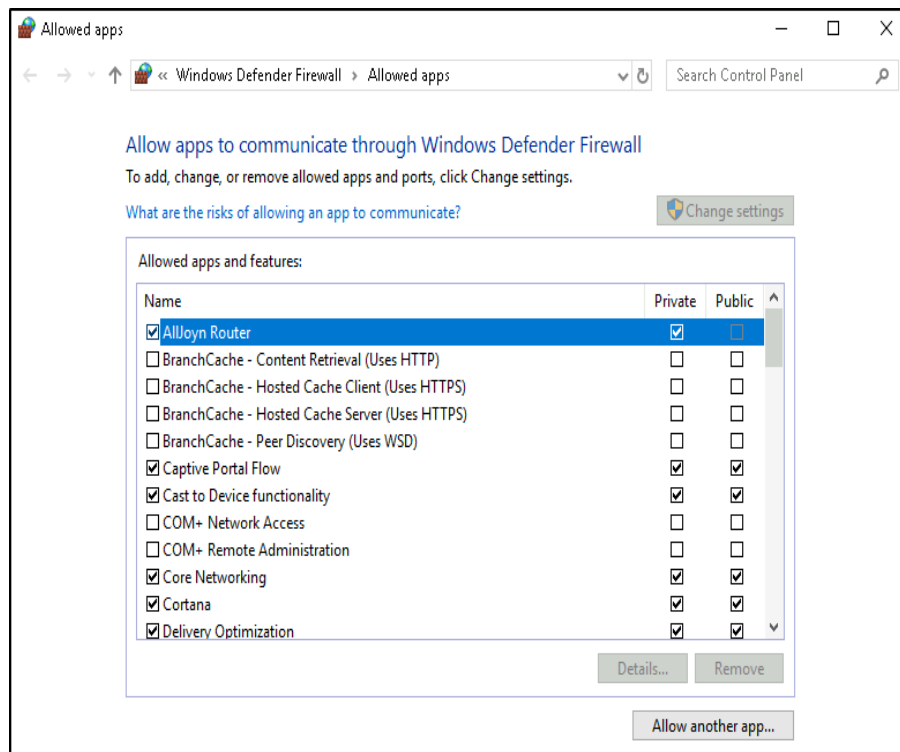
Click on any firewall profile, another screen will appear with two options: **turn the firewall on/off** and **block all incoming connections**.



Entry 8 [continued]

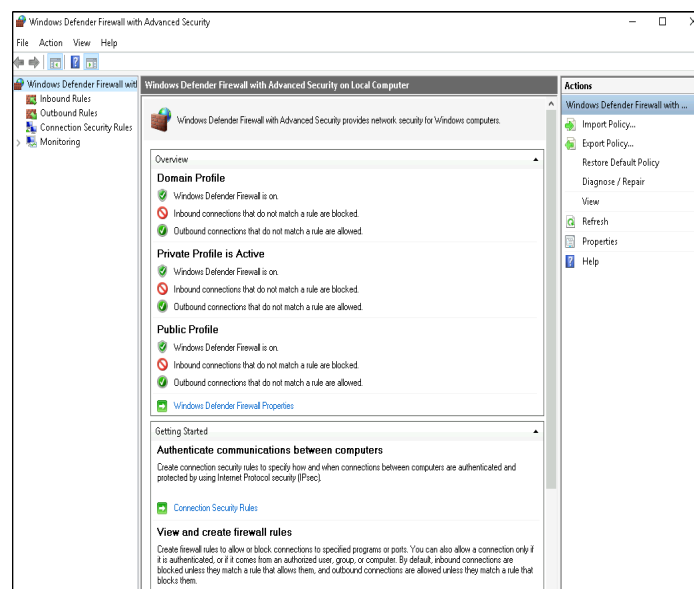
It is recommended to leave **Windows Defender Firewall** enabled.

Allow an app through firewall



View what the current settings are for any firewall profile as shown in the above image, several apps have access in the Private and/or Public firewall profile. Some of the apps will provide additional information if it's available via the **Details** button. Configuring the **Windows Defender Firewall** is for advanced Windows users. **Tip:** Command to open the Windows Defender Firewall is **WF.msc**.

Advanced Settings



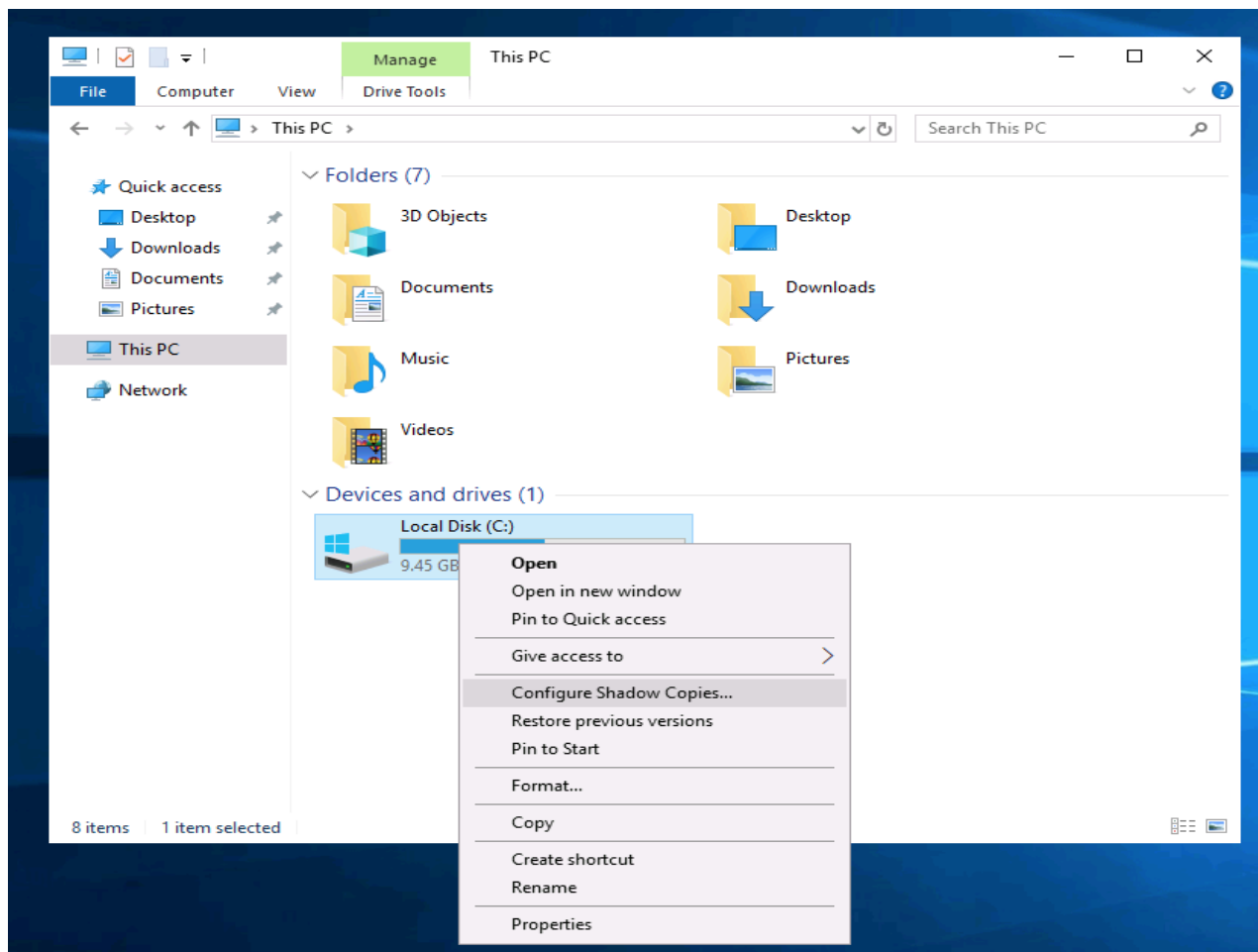
Entry 8 [continued]

Per Microsoft the Volume Shadow Copy Service (VSS) coordinates the required actions to create a consistent shadow copy (also known as a snapshot or a point-in-time copy) of the data that is to be backed up. Volume Shadow Copies are stored on the System Volume Information folder on each drive that has protection enabled.

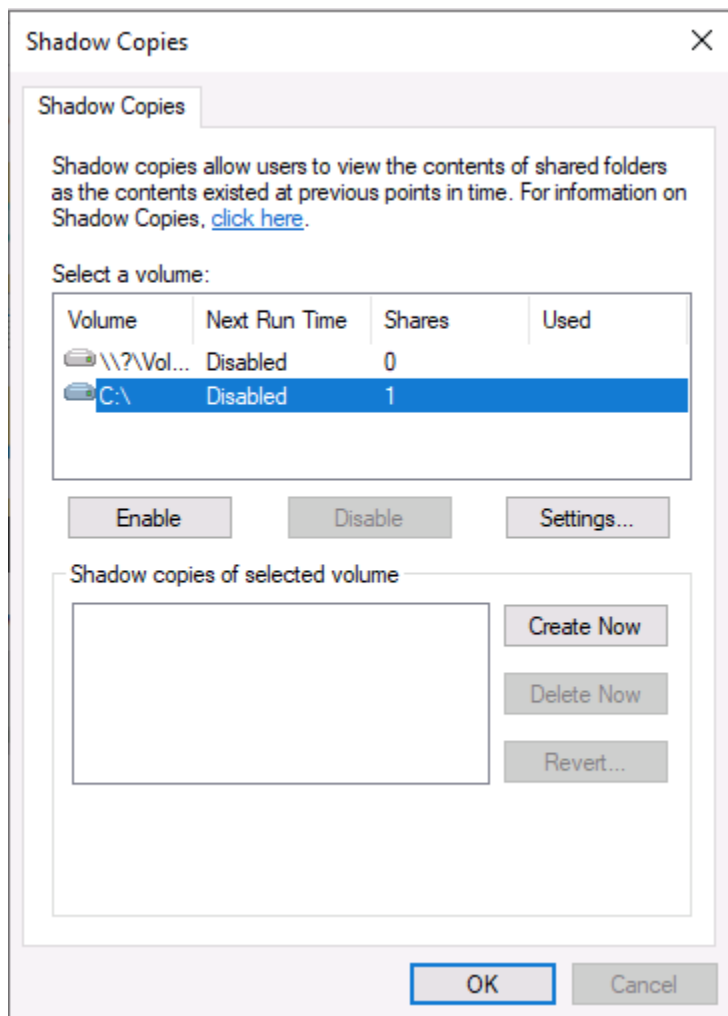
If VSS is enabled (**System Protection** turned on). Following tasks can be performed from within **advanced system settings**.

- **Create a restore point**
- **Perform system restore**
- **Configure restore settings**
- **Delete restore points**

From a security perspective, malware writers know of this Windows feature and write code in their malware to look for these files and delete them. Doing so makes it impossible to recover from a ransomware attack unless using an offline/off-site backup. To configure Shadow Copies within the attached VM, see below.



Entry 8 [continued]



Important Takeaways:

Covered was several built-in Windows security tools that ship with the Windows OS to help keep the device protected. There is much to explain and cover regarding the Windows OS. As mentioned, "The content is aimed at those who wish to understand and use the Windows OS on a more comfortable level."

Further reading material:

- [Antimalware Scan Interface](#)
- [Windows 10 Hello](#)
- [CSO Online - The best new Windows 10 security features](#)

Note: Attackers use built-in Windows tools and utilities in an attempt to go undetected within the victim environment. This tactic is known as Living Off The Land. Refer to the following resource [here](#) to learn more about this.

Entry 9

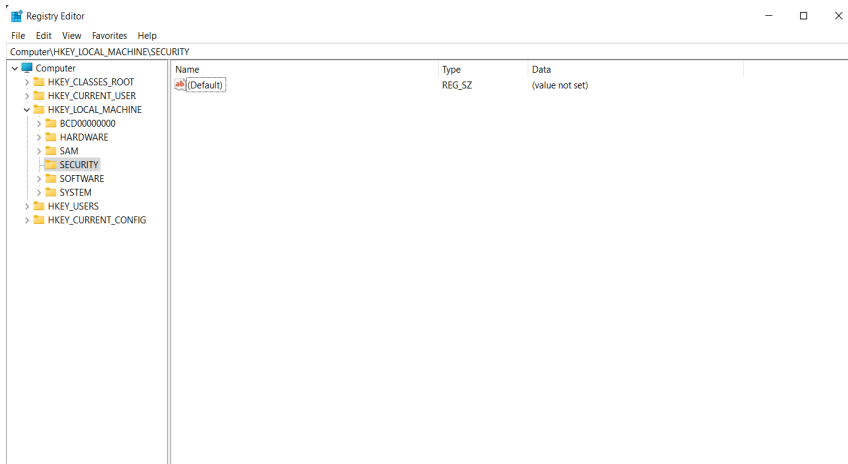
Room Name: Windows Forensics 1

Date Completed: 04/02/2026

Notes During the Room:

Windows Registry:

The Windows Registry is a collection of databases that contains the system's configuration data. This configuration data can be about the hardware, the software, or the user's information. It also includes data about the recently used files, programs used, or devices connected to the system. Understanding this data is beneficial from a forensics standpoint. The registry uses regedit.exe, a built-in Windows utility to view and edit the registry. The Windows registry consists of Keys and Values. When opening the regedit.exe utility to view the registry, the folders seen are Registry Keys. Registry Values are the data stored in these Registry Keys. A [Registry Hive](#) is a group of Keys, subkeys, and values stored in a single file on the disk.



Show above are root keys in the left pane in a tree view that shows the included registry keys, and the values in the selected key are shown in the right pane. Right-click on the value shown in the right pane and select properties to view the properties of this value. Microsoft defines each of these root keys. For more detail and information about the following Windows registry keys, please visit [Microsoft's documentation](#).

To access a live system, be able to access the registry using regedit.exe, and be greeted with all of the standard root keys we learned about in the previous task. However, if having access to a disk image, there must be a realization where the registry hives are located on the disk. The majority of these hives are located in the **C:\Windows\System32\Config** directory and are:

1. **DEFAULT** (mounted on **HKEY_USERS\DEFAULT**)
2. **SAM** (mounted on **HKEY_LOCAL_MACHINE\SAM**)
3. **SECURITY** (mounted on **HKEY_LOCAL_MACHINE\Security**)
4. **SOFTWARE** (mounted on **HKEY_LOCAL_MACHINE\Software**)
5. **SYSTEM** (mounted on **HKEY_LOCAL_MACHINE\System**)

Entry 9 [continued]

Hives containing user information: Apart from these hives, two other hives containing user information can be found in the User profile directory. For Windows 7 and above, a user's profile directory is located in **C:\Users\<username>** where the hives are:

1. **NTUSER.DAT** (mounted on HKEY_CURRENT_USER when a user logs in)
2. **USRCLASS.DAT** (mounted on HKEY_CURRENT_USER\Software\CLASSES)

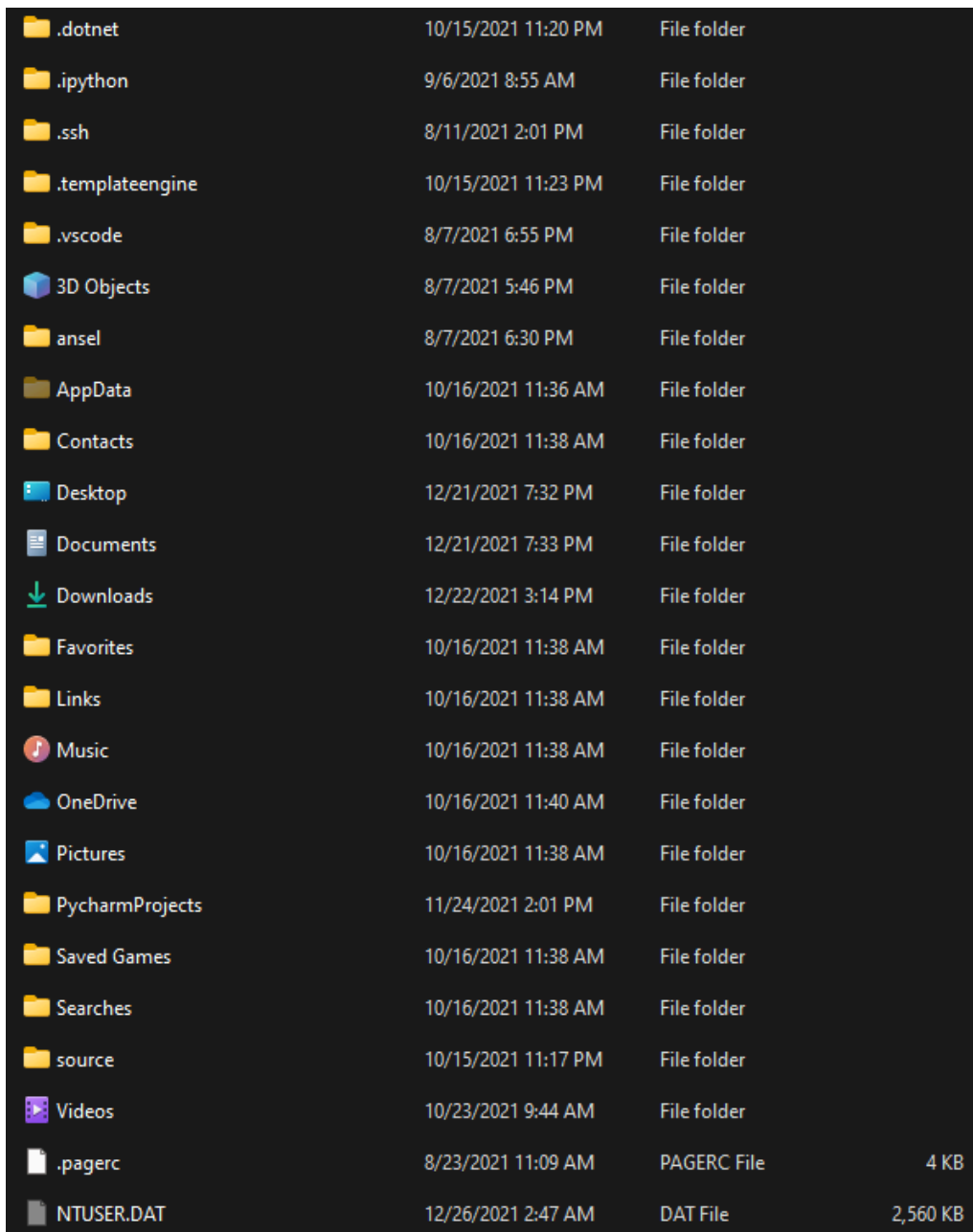
The USRCLASS.DAT hive is located in the directory

C:\Users\<username>\AppData\Local\Microsoft\Windows.

0	10/16/2021 11:38 AM	File folder	
1024	10/21/2021 10:56 AM	File folder	
1033	10/16/2021 11:38 AM	File folder	
ActionCenterCache	12/26/2021 2:47 AM	File folder	
Application Shortcuts	10/16/2021 11:38 AM	File folder	
Burn	10/16/2021 11:39 AM	File folder	
Caches	12/26/2021 11:40 AM	File folder	
CloudStore	6/5/2021 8:10 AM	File folder	
Explorer	10/23/2021 11:45 AM	File folder	
Fonts	9/3/2021 9:32 PM	File folder	
GameExplorer	6/5/2021 8:10 AM	File folder	
History	10/16/2021 11:36 AM	File folder	
Notifications	8/7/2021 5:46 PM	File folder	
PowerShell	10/21/2021 10:51 AM	File folder	
PPBCompatCache	10/16/2021 11:55 AM	File folder	
PPBCompatUaCache	10/16/2021 11:55 AM	File folder	
Ringtones	10/16/2021 11:38 AM	File folder	
RoamingTiles	8/7/2021 5:46 PM	File folder	
Safety	10/24/2021 10:12 PM	File folder	
Shell	9/13/2021 11:40 AM	File folder	
Themes	8/8/2021 7:36 PM	File folder	
WinX	12/7/2019 4:14 AM	File folder	
usrClass.dat	12/26/2021 2:47 AM	DAT File	5,120 KB

Entry 9 [continued]

The NTUSER.DAT hive is located in the directory `C:\Users\<username>\`.



.dotnet	10/15/2021 11:20 PM	File folder	
.python	9/6/2021 8:55 AM	File folder	
.ssh	8/11/2021 2:01 PM	File folder	
.templateengine	10/15/2021 11:23 PM	File folder	
.vscode	8/7/2021 6:55 PM	File folder	
3D Objects	8/7/2021 5:46 PM	File folder	
ansel	8/7/2021 6:30 PM	File folder	
AppData	10/16/2021 11:36 AM	File folder	
Contacts	10/16/2021 11:38 AM	File folder	
Desktop	12/21/2021 7:32 PM	File folder	
Documents	12/21/2021 7:33 PM	File folder	
Downloads	12/22/2021 3:14 PM	File folder	
Favorites	10/16/2021 11:38 AM	File folder	
Links	10/16/2021 11:38 AM	File folder	
Music	10/16/2021 11:38 AM	File folder	
OneDrive	10/16/2021 11:40 AM	File folder	
Pictures	10/16/2021 11:38 AM	File folder	
PycharmProjects	11/24/2021 2:01 PM	File folder	
Saved Games	10/16/2021 11:38 AM	File folder	
Searches	10/16/2021 11:38 AM	File folder	
source	10/15/2021 11:17 PM	File folder	
Videos	10/23/2021 9:44 AM	File folder	
.pagerc	8/23/2021 11:09 AM	PAGERC File	4 KB
NTUSER.DAT	12/26/2021 2:47 AM	DAT File	2,560 KB

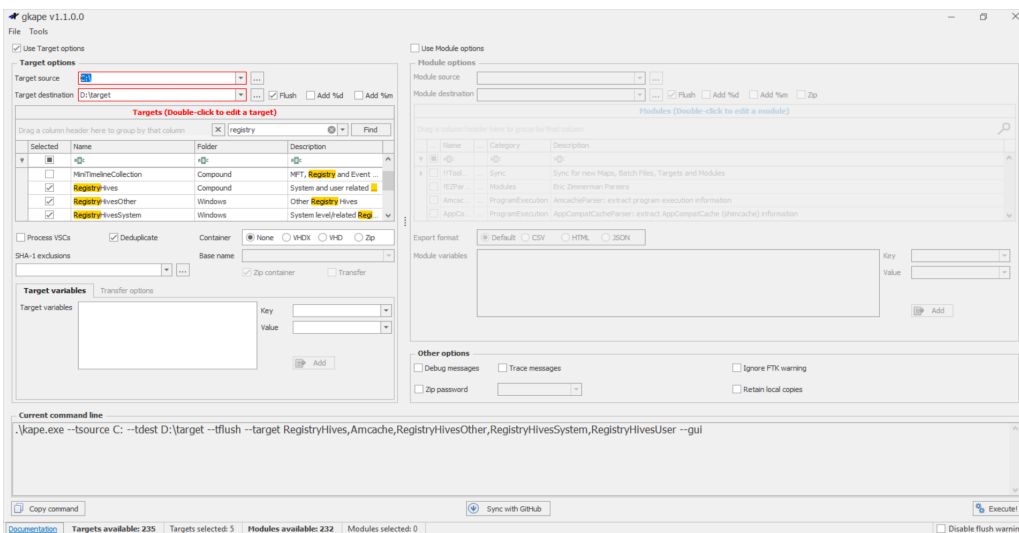
Remember that NTUSER.DAT and USRCLASS.DAT are hidden files.

The Amcache Hive: Apart from these files, there is another very important hive called the AmCache hive. This hive is located in `C:\Windows\AppCompat\Programs\Amcache.hve`. Windows creates this hive to save information on programs that were recently run on the system.

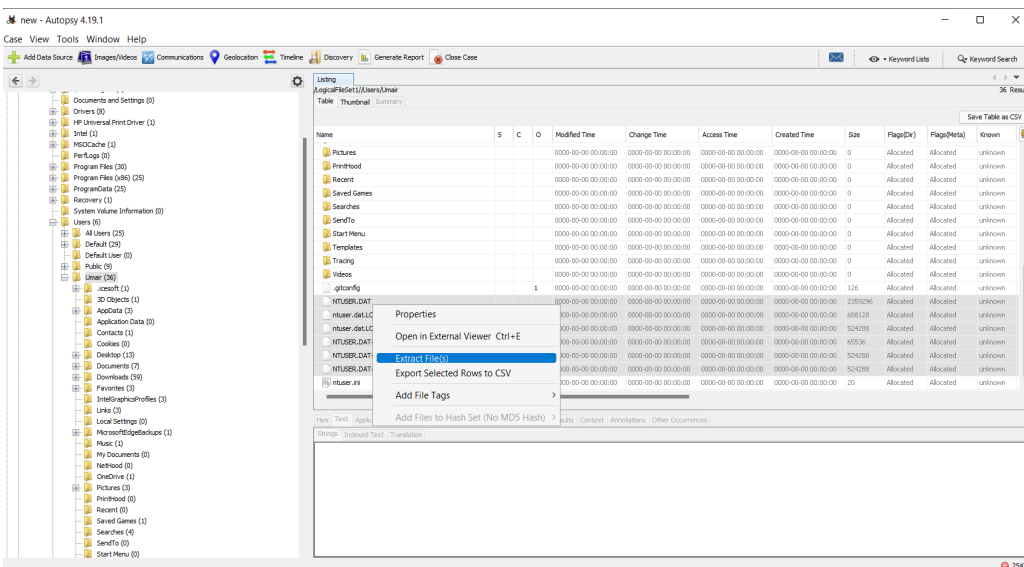
Entry 9 [continued]

When performing forensics, there is an encounter with a live system or an image taken of the system. For the sake of accuracy, it is recommended practice to image the system or make a copy of the required data and perform forensics on it. This process is called data acquisition. Though we can view the registry through the registry editor, the forensically correct method is to acquire a copy of this data and perform analysis on that. We can go to copy the registry hives from `%WINDIR%\System32\Config`, we cannot because it is a restricted file. **Acquiring these files, we can use one of the following tools:**

KAPE: ([KAPE](#)) is a live data acquisition and analysis tool which can be used to acquire registry data. It is primarily a command-line tool but also comes with a GUI. Screenshot below shows the KAPE GUI. We have already selected all the settings to extract the registry data using KAPE in this screenshot. We will learn more about collecting forensic artifacts using KAPE in a dedicated KAPE room.

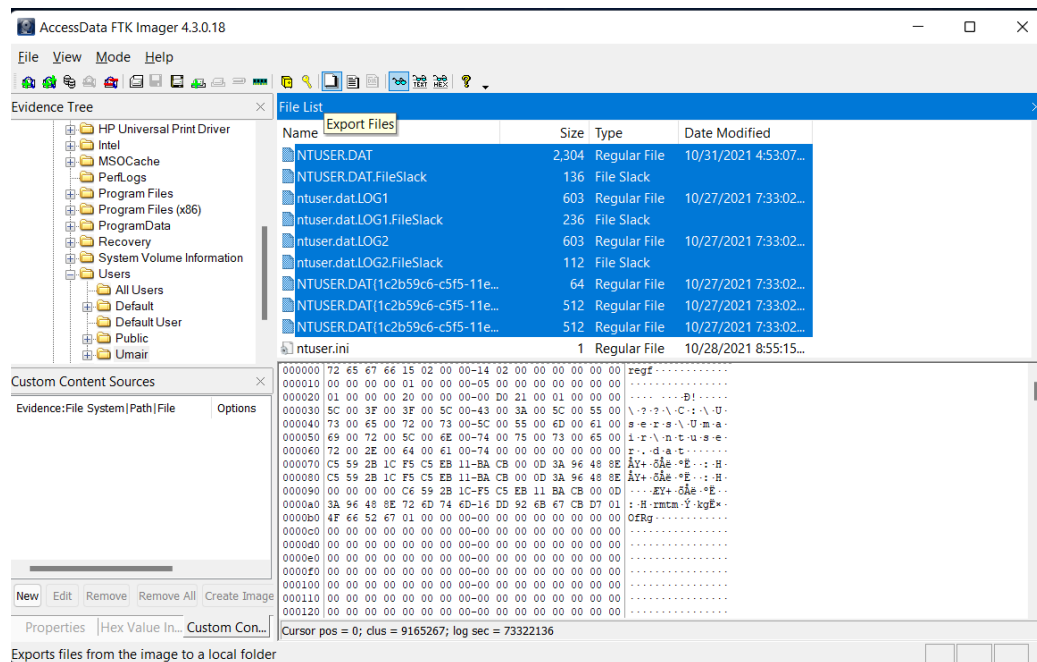


Autopsy: ([Autopsy](#)) gives the option to acquire data from both live systems or from a disk image. After adding data source, navigate to the location of the files to be extracted, then right-click and select the Extract File(s) option. It will look similar to what is in the screenshot below.

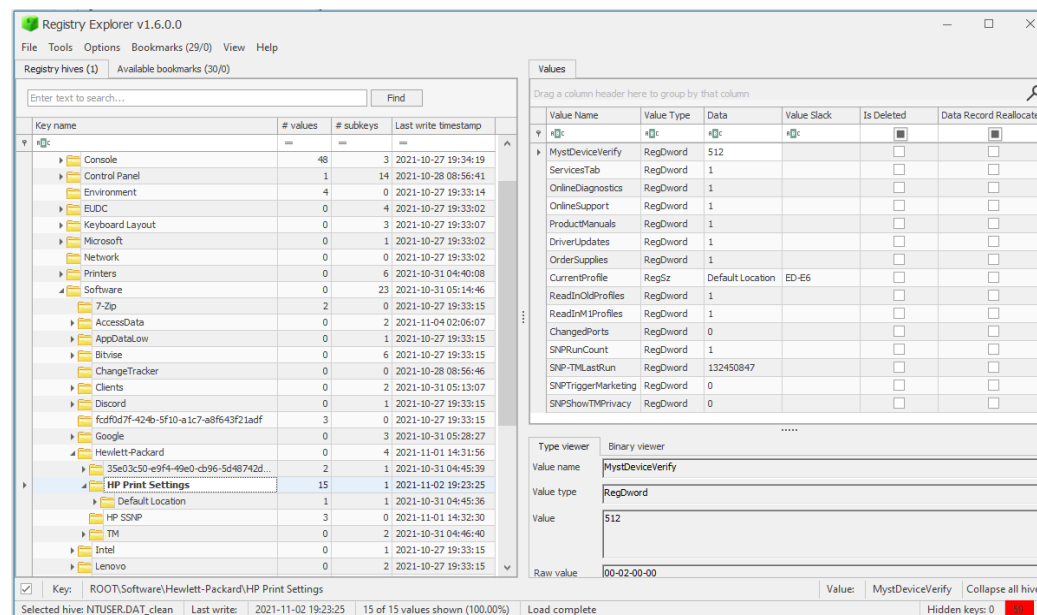


Entry 9 [continued]

FTK Imager: ([FTK Imager](#)) is similar to Autopsy and allows you to extract files from a disk image or a live system by mounting the said disk image or drive in FTK Imager. Below you can see the option to Export files as highlighted in the screenshot

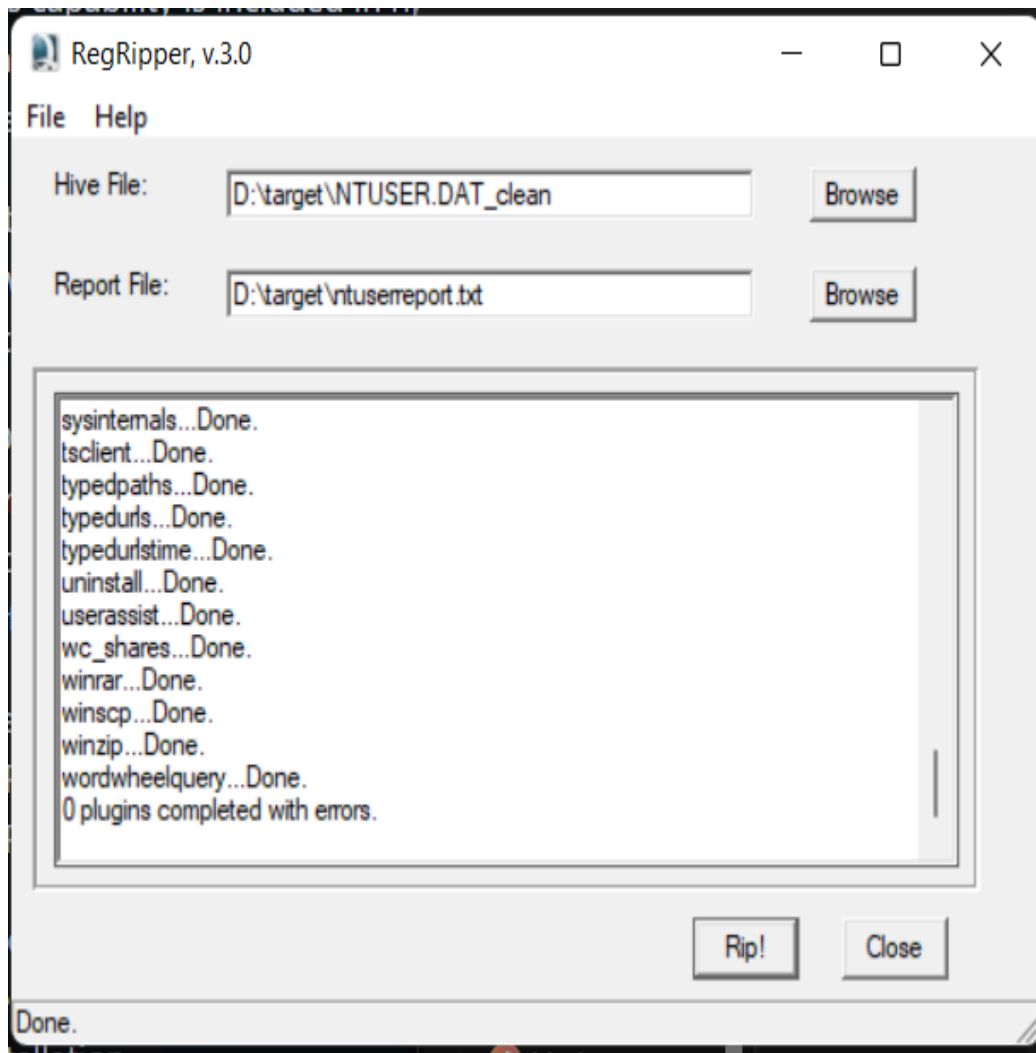


Zimmerman's Registry Explorer: Eric Zimmerman developed [tools](#) very useful for performing Digital Forensics and Incident Response. One of them is known as 'Registry Explorer'. It can load multiple hives simultaneously and add data from transaction logs into the hive to make a more 'cleaner' hive with more up-to-date data. It also has a handy 'Bookmarks' option containing forensically important registry keys often sought by forensics investigators who can go straight to the interesting registry keys and values with the bookmarks menu item.



Entry 9 [continued]

RegRipper: ([RegRipper](#)) is a utility that takes a registry hive as input and outputs a report that extracts data from some of the forensically important keys and values in that hive. The output report is in a text file and shows all the results in sequential order. RegRipper is available in both a CLI and GUI form which is shown in the screenshot below.



One shortcoming of RegRipper is that it does not take the transaction logs into account. The use of Registry Explorer to merge transaction logs with the respective registry hives before sending the output to RegRipper for a more accurate result.

Important Takeaways:

N / A

Entry 10

Room Name: Windows Forensics 2

Date Completed: 04/03/2026

Notes During the Room:

Windows Forensics practices extracting forensic artifacts from the Windows Registry by gathering system information, user information, files and folders accessed, programs run, and external devices connected to the system, all from the Windows registry. The Registry is not the only place where forensic artifacts are present. Different file systems are commonly used by Windows where to look for artifacts and identify locations and artifacts to prove evidence of execution, file/folder usage or knowledge, and external device usage as well as basic recovering deleted files. The use of [Eric Zimmerman's tools](#) parse information present in the artifacts as we already used Registry Explorer and ShellBags Explorer and for this task, we will use Autopsy.

The File Allocation Table (FAT): The File Allocation Table (FAT) is the default file system for Microsoft Operating Systems. FAT creates a table that indexes the location of bits allocated to different files.

Data structures of the FAT file system: The FAT file system supports the following Data structures.

Cluster: A cluster is a basic storage unit of the FAT file system. Each file stored on a storage device can be considered a group of clusters containing bits of information.

Directory: A directory contains information about file identification, like file name, starting cluster, and filename length.

File Allocation Table: The File Allocation Table is a linked list of all the clusters. It contains the status of the cluster and the pointer to the next cluster in the chain. The bits that make up a file are stored in clusters. All the filenames on a file system, their starting clusters, and their lengths are stored in directories. And the location of each cluster on the disk is stored in the File Allocation Table. We can see that we started with a raw disk composed of bits and organized it to define what group of bits refers to what file stored on the disk. The FAT file format divides the available disk space into clusters for more straightforward addressing. The number of these clusters depends on the number of bits used to address the cluster. Hence the different variations of the FAT file system. FAT was initially developed with 8-bit cluster addressing, and it was called the FAT Structure. Later, as the storage needed to be increased, FAT12, FAT16, and FAT32 were introduced. Even though the maximum volume size for FAT32 is 2TB, Windows limits formatting to only 32GB. However, volume sizes formatted on other OS with larger volume sizes are supported by Windows. The chances of coming across a FAT12 filesystem are very rare nowadays. FAT16 and FAT32 are still used in some places, like USB drives, SD cards, or Digital cameras. However, the maximum file size and volume size (4GB - 1 file size for both FAT16 and FAT32) are limiting factors that have reduced their usage.

Entry 10 [continued]

Attribute	FAT12	FAT16	FAT32
Addressable bits	12	16	28
Max number of clusters	4,096	65,536	268,435,456
Supported size of clusters	512B - 8KB	2KB - 32KB	4KB - 32KB
Maximum Volume size	32MB	2GB	2TB



The exFAT file system: As the file sizes have grown, especially with higher resolution images and videos being supported by the newer digital cameras, the maximum file size limit of FAT32 became a substantial limiting factor for camera manufacturers. Though Microsoft had moved on to the NTFS file system, it was not suitable for digital media devices as they did not need the added security features and the overhead that came with it. The exFAT file system is now the default for SD cards larger than 32GB. The exFAT file system supports a cluster size of 4KB to 32MB. It has a maximum file size and a maximum volume size of 128PB (Petabytes). It can have a maximum of 2,796,202 files per directory.

File system: As observed in the previous task, the FAT file system is a very basic file system. It does the job when it comes to organizing our data, but it offers little more in terms of security, reliability, and recovery capabilities. It also has certain limitations when it comes to file and volume sizes. Hence, Microsoft developed a newer file system called the New Technology File System (NTFS) to add these features. This file system was introduced in 1993 with Windows NT 3.1. However, it became mainstream after Windows XP. The NTFS file system resolves many issues present in the FAT file system and introduces a lot of new features. We will discuss some of the features below.

Journaling: The NTFS file system keeps a log of changes to the metadata in the volume. This feature helps the system recover from a crash or data movement due to defragmentation. This log is stored in \$LOGFILE in the volume's root directory. Hence the NTFS file system is called a journaling file system.

Access Controls: The FAT file system did not have access controls based on the user. The NTFS file system has access controls that define the owner of a file/directory and permissions for each user.

Entry 10 [continued]

Volume Shadow Copy: The NTFS file system keeps track of changes made to a file using a feature called Volume Shadow Copies. Using this feature, a user can restore previous file versions for recovery or system restore. In recent ransomware attacks, ransomware actors have been noted to delete the shadow copies on a victim's file systems to prevent them from recovering their data.

Alternate Data Streams: A file is a stream of data organized in a file system. Alternate data streams (ADS) is a feature in NTFS that allows files to have multiple streams of data stored in a single file. Internet Explorer and other browsers use Alternate Data Streams to identify files downloaded from the internet (using the ADS Zone Identifier). Malware has also been observed to hide their code in ADS.

Master File Table: Like the File Allocation Table, there is a Master File Table in NTFS. However, the Master File Table, or MFT, is much more extensive than the File Allocation Table. It is a structured database that tracks the objects stored in a volume. Therefore, we can say that the NTFS file system data is organized in the Master File Table. From a forensics point of view, the following are some of the critical files in the MFT:

\$MFT: The \$MFT is the first record in the volume. The Volume Boot Record (VBR) points to the cluster where it is located. \$MFT stores information about the clusters where all other objects present on the volume are located. This file contains a directory of all the files present on the volume.

\$LOGFILE: The \$LOGFILE stores the transactional logging of the file system. It helps maintain the integrity of the file system in the event of a crash.

\$UsnJrnl: It stands for the Update Sequence Number (USN) Journal. It is present in the \$Extend record. It contains information about all the files that were changed in the file system and the reason for the change. It is also called the change journal.

Deleted files and Data recovery: File systems make it easier to know how files are deleted, recovered, and wiped. A file system stores the location of a file on the disk in a table or a database. When deleted a file from the file system deletes the entries that store the file's location on the disk. The location where the file existed is now available for writing or unallocated. However, the file contents on disk are still there, as long as they are not overwritten by the file system while copying another file or by the disk firmware while performing maintenance on the disk. Similarly, there is data on the disk in different unallocated clusters, which can possibly be recovered. To recover this data, we have to understand the file structure of different file types to identify the specific file through the data we see in a hex editor.

Disk image: A disk image file is a file that contains a bit-by-bit copy of a disk drive. A bit-by-bit copy saves all the data in a disk image file, including the metadata, in a single file. Thus, while performing forensics, one can make several copies of the physical evidence, i.e., the disk, and use them for investigation. This helps in two ways. 1) The original evidence is not contaminated while performing forensics, and 2) The disk image file can be copied to another disk and analyzed without using specialized hardware.

Important Takeaways: [Windows Forensics 1](#) learning was Windows registry and different artifacts.

Entry 11

Room Name: Intro to Log Analysis

Date Completed: 04/07/2026

Notes During the Room:

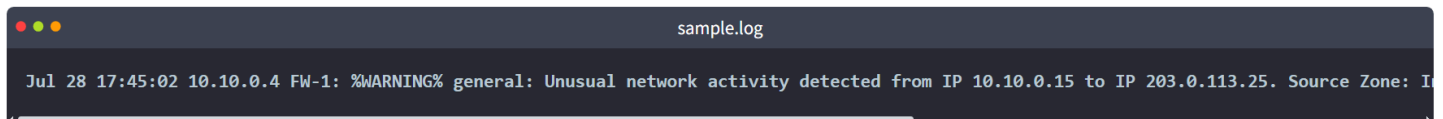
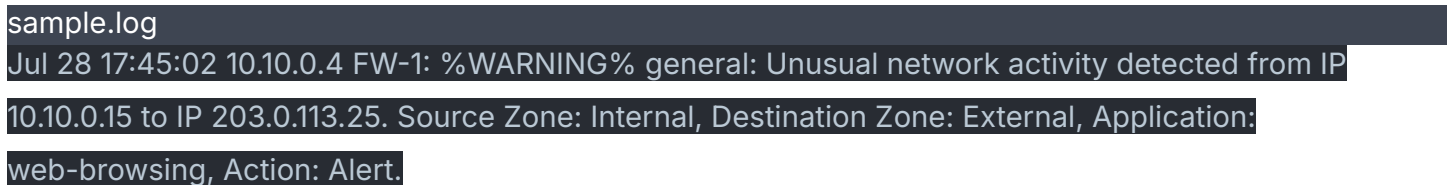
Log analysis is an essential aspect of cyber security and system monitoring. At a high level, log analysis examines and interprets log event data generated by various sources (devices, applications, and systems) to monitor metrics and identify security incidents. It involves collecting, parsing, and processing log files to turn data into actionable objectives. By adopting an effective log analysis strategy, security teams can more accurately respond to security incidents and gain proactive insights into potential threats. Concepts related to log analysis methodology, effective logging practices, and common tools to aid detection and response.

Learning Objectives

- Learn log analysis best practices.
- Discover the essential tools for log analysis.
- Gain hands-on experience in analyzing logs by using multiple tools and technologies.

What Are Logs?

Logs are recorded events or transactions within a system, device, or application. Specifically, these events can be related to application errors, system faults, audited user actions, resource uses, network connections, and more. Each log entry contains relevant details to contextualize the event, such as its timestamp (the date and time it occurred), the source (the system that generated the log), and additional information about the specific log event.

A screenshot of a terminal window with a dark background. The title bar at the top says "sample.log". Inside the terminal, there is a single line of text representing a log entry: "Jul 28 17:45:02 10.10.0.4 FW-1: %WARNING% general: Unusual network activity detected from IP 10.10.0.15 to IP 203.0.113.25. Source Zone: I".A text representation of the log entry shown in the screenshot above, with line breaks. It reads: "sample.log", "Jul 28 17:45:02 10.10.0.4 FW-1: %WARNING% general: Unusual network activity detected from IP", "10.10.0.15 to IP 203.0.113.25. Source Zone: Internal, Destination Zone: External, Application:", "web-browsing, Action: Alert.".

In the above example, this log entry signifies an event detected by a firewall regarding unusual network activity from an internal system, indicating a potential security concern.

Entry 11 [continued]

The relevant fields to consider in this example are:

Jul 28 17:45:02 - This timestamp shows the event's date and time.

10.10.0.4 - This refers to the system's IP address (the source) that generated the log.

%WARNING% - This indicates the severity of the log, in this case, Warning. Log entries are often given a severity level to categorize and communicate their relative importance or impact. These severity levels help prioritize responses, investigations, and actions based on the criticality of the events. Different systems might use slightly different severity levels, but commonly, you can expect to find the following increasing severity levels: **Informational**, **Warning**, **Error**, and **Critical**.

Action: Alert - Firewall's policy was configured to notify when such unusual activity occurs.

The remaining fields give us specific information related to the logged event. Specifically, that unusual network activity was detected from IP 10.10.0.15 to IP 203.0.113.25. Based on the **Source Zone** field, the traffic appears destined for the Internet (**External**), and the **Application** was categorized as web-browsing.

Why Are Logs Important?

There are several reasons why collecting logs and adopting an effective log analysis strategy is vital for an organization's ongoing operations. Some of the most common activities include:

- **System Troubleshooting:** Analyzing system errors and warning logs helps IT teams understand and quickly respond to system failures, minimizing downtime, and improving overall system reliability.
- **Cyber Security Incidents:** In the security context, logs are crucial in detecting and responding to security incidents. Firewall logs, intrusion detection system (IDS) logs, and system authentication logs, for example, contain vital information about potential threats and suspicious activities. Performing log analysis helps SOC teams and Security Analysts identify and quickly respond to unauthorized access attempts, malware, data breaches, and other malicious activities.
- **Threat Hunting:** On the proactive side, cyber security teams can use collected logs to actively search for advanced threats that may have evaded traditional security measures. Security Analysts and Threat Hunters can analyze logs to look for unusual patterns, anomalies, and indicators of compromise (IOCs) that might indicate the presence of a threat actor.
- **Compliance:** Organizations must often maintain detailed records of their system's activities for regulatory and compliance purposes. Regular log analysis ensures that organizations can provide accurate reports and demonstrate compliance with regulations such as GDPR, HIPAA, or PCI DSS.

Entry 11 [continued]

Common Log File Locations

A crucial aspect of log analysis is understanding where to locate log files generated by various applications and systems. While log file paths can vary due to system configurations, software versions, and custom settings, knowing common log file locations is essential for efficient investigation and threat detection.

- **Web Servers:**
 - **Nginx:**
 - Access Logs: `/var/log/nginx/access.log`
 - Error Logs: `/var/log/nginx/error.log`
 - **Apache:**
 - Access Logs: `/var/log/apache2/access.log`
 - Error Logs: `/var/log/apache2/error.log`
- **Databases:**
 - **MySQL:**
 - Error Logs: `/var/log/mysql/error.log`
 - **PostgreSQL:**
 - Error and Activity Logs: `/var/log/postgresql/postgresql-{version}-main.log`
- **Web Applications:**
 - **PHP:**
 - Error Logs: `/var/log/php/error.log`
- **Operating Systems:**
 - **Linux:**
 - General System Logs: `/var/log/syslog`
 - Authentication Logs: `/var/log/auth.log`
- **Firewalls and IDS/IPS:**
 - **iptables:**
 - Firewall Logs: `/var/log/iptables.log`
 - **Snort:**
 - Snort Logs: `/var/log/snort/`

While these are common log file paths, it's important to note that actual paths may differ based on system configurations, software versions, and custom settings. It's recommended to consult the official documentation or configuration files to verify the correct log file paths to ensure accurate analysis and investigation.

Entry 11 [continued]

Automated Analysis

Automated analysis involves the use of tools. For example, these often include commercial tools such as XPLG or SolarWinds Loggly. Automated analysis tools allow for processing and data analysis of logs. These tools often utilize Artificial Intelligence / Machine Learning to analyze patterns and trends. As the AI landscape evolves, we expect to see more effective automated analysis solutions.

Advantages	Disadvantages
Saves time by performing a lot of the manual work required in manual analysis	Automated analysis tools are usually commercial-only and, therefore, expensive.
The use of artificial intelligence is effective at recognizing patterns and trends.	The effectiveness of artificial intelligence depends on how capable the model is. For example, the risk of false positives increases, or newer or never-seen-before events can be missed as the AI is not trained to recognize these.

Manual Analysis

Manual analysis is the process of examining data and artifacts without using automation tools. For example, an analyst scrolling through a web server log would be considered manual analysis. Manual analysis is essential for an analyst because automation tools cannot be relied upon.

Advantages	Disadvantages
It is cheap and does not require expensive tooling. For example, simple Linux commands can do the trick.	It is time-consuming as the analyst has to do all of the work, including reformatting log files.
Allows for a thorough investigation.	N/A
Reduces the risk of overfitting or false positives on alerts from automated tools.	Events or alerts can be missed! Especially if there is a lot of data to comb through.
Allows for contextual analysis. The analyst has a broader understanding of the organization and cyber security landscape.	N/A

Important Takeaways: Covered the basic methodology behind adopting an effective log analysis strategy as well as explored the importance of log data collection, common attack patterns, and useful tools for the investigation and response processes. Covered the basic methodology behind adopting an effective log analysis strategy. Explored the importance of log data collection, common attack patterns, and useful tools for the investigation and response processes.

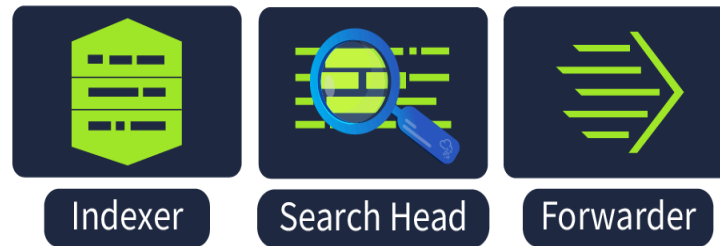
Entry 12

Room Name: Splunk Basics

Date Completed: 04/10/2026

Notes During the Room:

Splunk has three main components: Forwarder, Indexer, and Search Head. These components work together to help us search and analyze the data. These components are explained below:

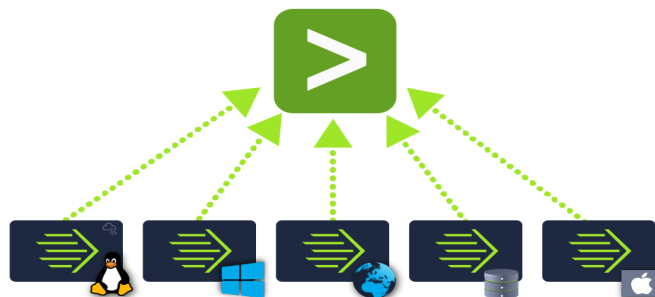


Splunk Forwarder

Splunk Forwarder is a lightweight agent installed on the endpoint intended to be monitored, and its main task is to collect the data and send it to the Splunk instance. It does not affect the endpoint's performance as it takes a few resources to process. Some of the key data sources are:

- Web server generating web traffic.
- Windows machine generating Windows Event Logs, PowerShell, and Sysmon data.
- Linux host generating host-centric logs.
- Database generating DB connection requests.

The forwarder collects the data from the log sources and sends it to the Splunk Indexer.



Entry 12 [continued]

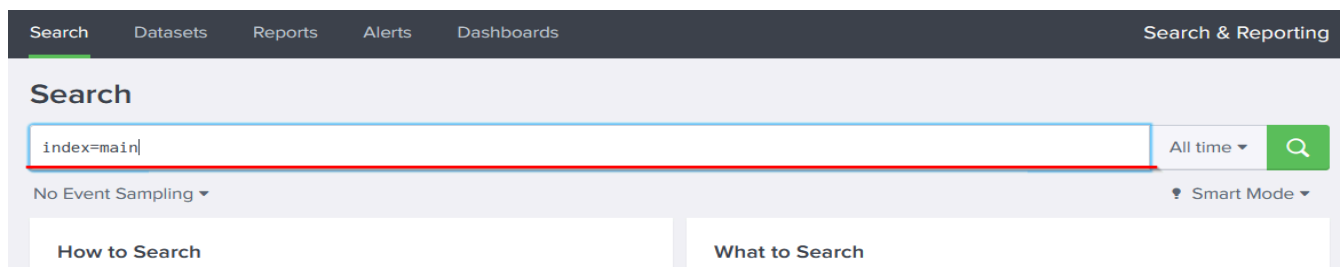
Splunk Indexer

Splunk Indexer plays the main role in processing the data it receives from forwarders. It parses and normalizes the data into field-value pairs, categorizes it, and stores the results as events, making the processed data easy to search and analyze.

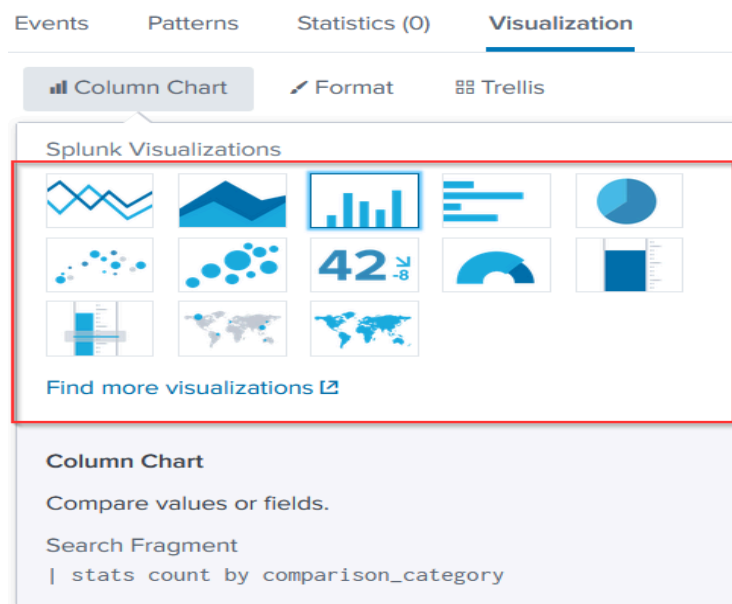


Search Head

Splunk Search Head is the place within the **Search & Reporting App** where users can search the indexed logs, as shown below. The searches are done using the SPL (Search Processing Language), a powerful query language for searching indexed data. When the user performs a search, the request is sent to the indexer, and the relevant events are returned as field-value pairs.



The Search Head also allows you to transform results into presentable tables and visualizations such as pie, bar, and column charts, as shown below:



Entry 12 [continued]

Splunk can ingest any data. According to the Splunk documentation, when data is added to Splunk, the data is processed and transformed into a series of individual events. The data sources can be event logs, website logs, firewall logs, etc. The data sources are grouped into categories. Below is a chart listing from the Splunk documentation detailing each data source category.

Data source	Description
Files and directories	Most data that you might be interested in comes directly from files and directories.
Network events	The Splunk software can index remote data from any network port and SNMP events from remote devices.
IT Operations	Data from IT Ops, such as Nagios, NetApp, and Cisco.
Cloud services	Data from Cloud services, such as AWS and Kinesis.
Database services	Data from databases such as Oracle, MySQL, and Microsoft SQL Server.
Security services	Data from security services such as McAfee, Microsoft Active Directory, and Symantec Endpoint Protection.
Virtualization services	Data from virtualization services such as VMWare and XenApp.
Application servers	Data from application servers such as JMX & JMS, WebLogic, and WebSphere.
Windows sources	The Windows version of Splunk software accepts a wide range of Windows-specific inputs, including Windows Event Log, Windows Registry, WMI, Active Directory, and Performance monitoring.
Other sources	Other input sources are supported, such as FIFO queues and scripted inputs for getting data from APIs, and other remote data interfaces.

In this task, we're going to focus on **VPN logs**. We're presented with the following screen when we click on the **Add Data** link on the Splunk home screen.

What data do you want to send to the Splunk platform?

Follow guides for onboarding popular data sources

**Cloud computing**
Get your cloud computing data in to the Splunk platform.
10 data sources

**Networking**
Get your networking data in to the Splunk platform.
2 data sources

**Operating System**
Get your operating system data in to the Splunk platform.
1 data source

**Security**
Get your security data in to the Splunk platform.
3 data sources

4 data sources in total

Or get data in with the following methods

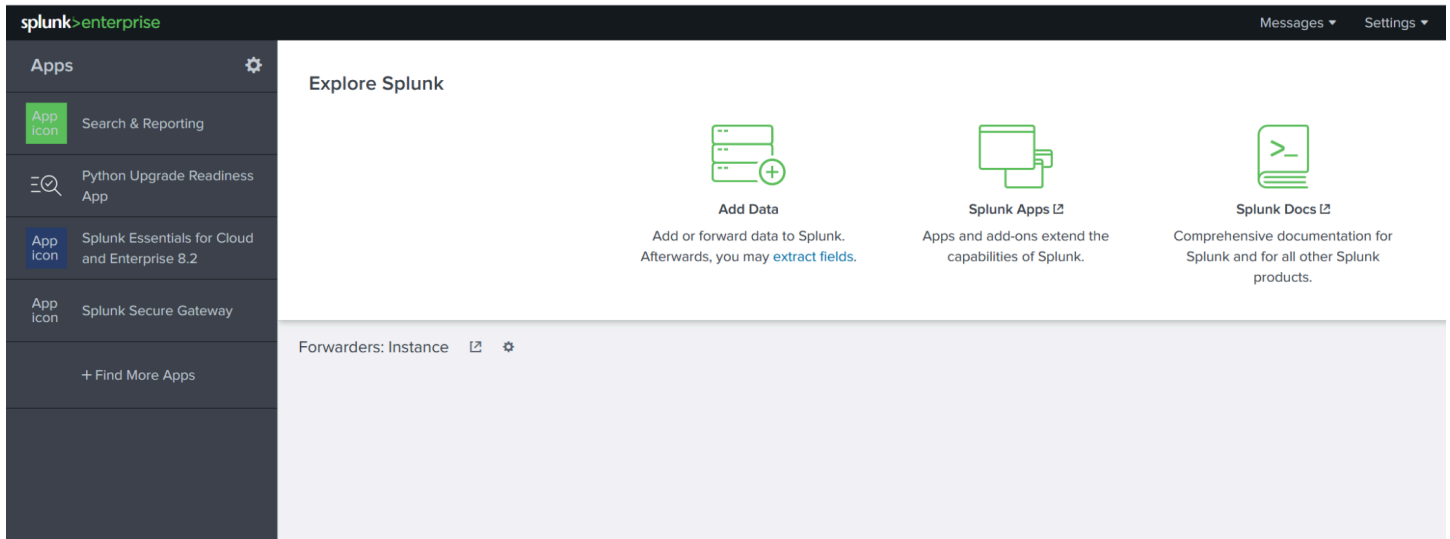
**Upload**
files from my computer
Local log files
Local structured files (e.g. CSV)
[Tutorial for adding data](#)

**Monitor**
files and ports on this Splunk platform instance
Files - HTTP - WMI - TCP/UDP - Scripts
Modular inputs for external data sources

**Forward**
data from a Splunk forwarder
Files - TCP/UDP - Scripts

Entry 12 [continued]

When you access Splunk, you will see the default **home screen** as shown below:



Splunk Bar

The top panel is the **Splunk Bar** as shown below:



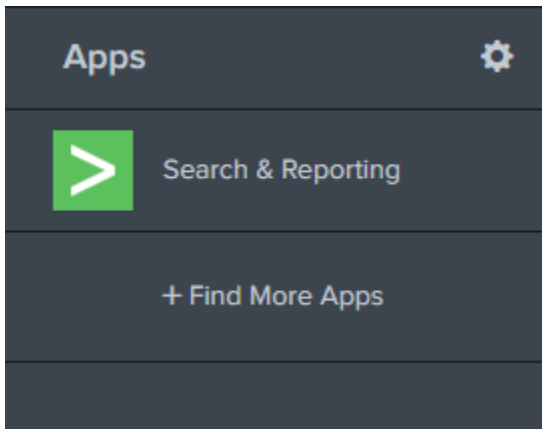
In the Splunk Bar, we have the following options available:

- **Messages:** View system-level notifications and messages.
- **Settings:** Configure Splunk instance settings.
- **Activity:** Review the progress of search jobs and processes.
- **Help:** View tutorials and documentation.
- **Find:** Search across the App.

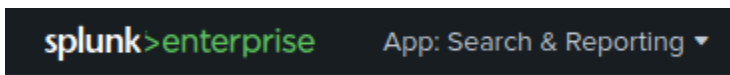
Entry 12 [continued]

Apps Panel

Next is the **Apps Panel**. This panel shows the apps installed for the Splunk instance. The default app for every Splunk installation is **Search & Reporting**.



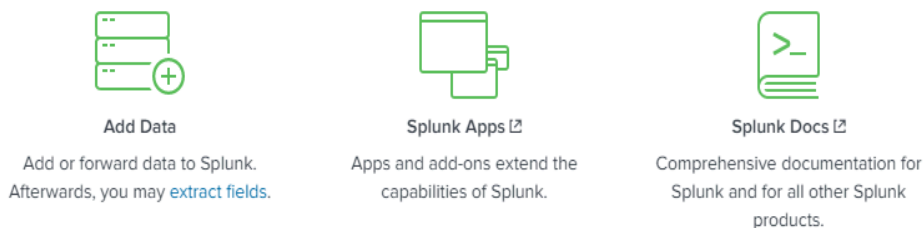
You can also switch between the Splunk Apps directly from the Splunk Bar, as shown below, without using the Apps Panel.



Explore Splunk

The next section is **Explore Splunk**. This panel contains quick links to add data to the Splunk instance, add new Splunk apps, and access the Splunk documentation.

Explore Splunk



Entry 12 [continued]

Important Takeaways:

Well done! In this room, you learned about [Splunk](#)'s core components, explored the [Splunk](#) interface, and practiced uploading data to [Splunk](#). You have gained the foundational knowledge of [Splunk SIEM](#).

If you'd like to dig deeper, you can explore the following [Splunk](#) walkthrough and challenge rooms to understand how [Splunk](#) is effectively used in investigating incidents.

- [Splunk: Exploring SPL](#)
- [Incident Handling with Splunk](#)
- [Investigating With Splunk](#)
- [Benign - Challenge](#)
- [PoshEclipse - Challenge](#)

Entry 13

Room Name: Incident Handling with Splunk

Date Completed: 04/13/2026

Notes During the Room:

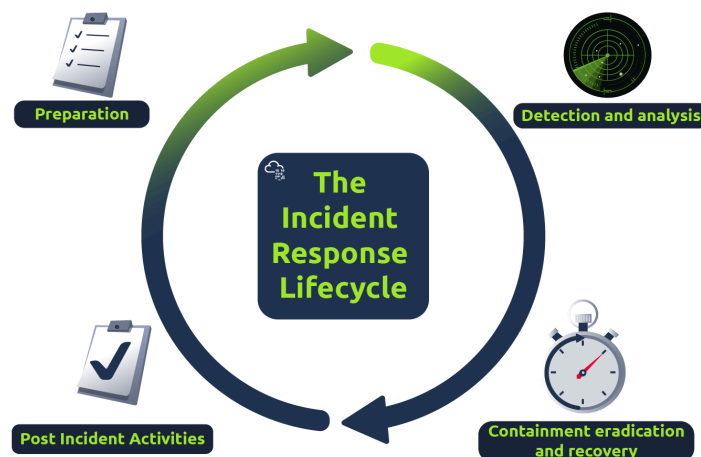
As an Incident Handler / SOC Analyst, we would aim to know the attackers' tactics, techniques, and procedures. Then we can stop/defend/prevent against the attack in a better way. The Incident Handling process is divided into four different phases. Let's briefly go through each phase before jumping into the incident, which we will be going through in this exercise.

1. Preparation: The preparation phase covers the readiness of an organization against an attack. That means documenting the requirements, defining the policies, incorporating the security controls to monitor like EDR / SIEM / IDS / IPS, etc. It also includes hiring/training the staff.

2. Detection and Analysis: The detection phase covers everything related to detecting an incident and the analysis process of the incident. This phase covers getting alerts from the security controls like SIEM/EDR investigating the alert to find the root cause. This phase also covers hunting for the unknown threat within the organization.

3. Containment, Eradication, and Recovery: This phase covers the actions needed to prevent the incident from spreading and securing the network. It involves steps taken to avoid an attack from spreading into the network, isolating the infected host, clearing the network from the infection traces, and gaining control back from the attack.

4. Post-Incident Activity / Lessons Learnt: This phase includes identifying the loopholes in the organization's security posture, which led to an intrusion, and improving so that the attack does not happen next time. The steps involve identifying weaknesses that led to the attack, adding detection rules so that similar breach does not happen again, and most importantly, training the staff if required.



Entry 13 [continued]

To investigate a cyber attack a Security Analysis is to investigate cyber attack and map the attacker's activities into 7 Cyber Kill Chain Phases. It is important to note no need to follow a sequence of the cyber kill chain. One finding in one phase will lead to another finding to map into some other phase.

Cyber Kill Chain

Follow the Cyber kill Chain Model map the attacker's activity in each phase during this Investigation. When required utilize Open Source Intelligence (OSINT) and other findings to fill the gaps in the kill chain. It is not necessary to follow this sequence of the phases while investigating.

- Reconnaissance
- Weaponization
- Delivery
- Exploitation
- Installation
- Command & Control
- Actions on Objectives

Important Takeaways:

Conclusion: As a SOC Analyst, an cyber-attack was investigated where the attacker had defaced a website 'imreallynotbatman.com' of the Wayne Enterprise. The attacker mapped activities went into the 7 phases of the Cyber Kill Chain.

Reconnaissance Phase: First look was into any reconnaissance activity from the attacker to identify the IP address and other details about the adversary.

Findings:

- IP Address **40.80.148.42** was found to be scanning our webserver.
- The attacker was using Acunetix as a web scanner.

Exploitation Phase: Looked into the traces of exploitation attempts and found brute-force attacks against our server, which were successful.

Findings:

- Brute force attack originated from IP **23.22.63.114.**
- The IP address used to gain access: **40.80.148.42**
- 142 unique brute force attempts were made against the server, out of which one attempt was successful

Entry 13 [continued]

Installation Phase: Next, look at the installation phase to see any executable from the attacker's IP Address uploaded to our server.

Findings:

- A malicious executable file `3791.exe` was observed to be uploaded by the attacker.
- Looked at the sysmon logs and found the MD5 hash of the file.

Action on Objective: After compromising the web server, the attacker defaced the website.

Findings:

- Examined the logs and found the file name used to deface the webserver.

Weaponization Phase: We used various threat Intel platforms to find the attacker's infrastructure based on the following information we saw in the above activities.

Information we had: Domain: `prankglassinebracket.jumpingcrab.com`

IP Address: `23.22.63.114`

Findings:

- Multiple masquerading domains were found associated with the attacker's IPs.
- An email of the user `Lillian.rose@po1s0n1vy.com` was also found associated with the attacker's IP address.

Deliver Phase: In this phase, we again leveraged online Threat Intel sites to find malware associated with the adversary's IP address, which appeared to be a secondary attack vector if the initial compromise failed.

Findings:

- A malware name: `MirandaTateScreensaver.scr.exe` was found associated with the adversary.
- MD5 of the malware was: `c99131e0169171935c5ac32615ed6261`

Entry 14

Room Name: Splunk 2

Date Completed:

Notes During the Room:

TryHackMe System - Inop

Important Takeaways:

Entry 15

Room Name: Splunk 3

Date Completed:

Notes During the Room:

TryHackMe System - Inop

Important Takeaways: